
When Shift Happens - Confounding Is to Blame

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Abstract

Distribution shifts inject uncertainty that erodes the robustness and generalization of machine learning models. Conventional wisdom holds that learning causal-invariant representations is the key to robustness under such shifts, yet recent empirical studies expose a striking counterexample: (i) empirical risk minimization (ERM) consistently rivals and often outperforms state-of-the-art out-of-distribution (OOD) generalization methods, and (ii) its OOD generalization performance improves when all available covariates—not just causal ones—are used. We identify the cause of this phenomenon: hidden confounding. Shifts in hidden confounding reshape data distributions in ways that directly violate the assumptions on which existing OOD generalization approaches rest. Under these conditions, we prove that effective generalization demands learning environment-specific relationships; relying on invariant relationships alone is provably insufficient. We further establish that augmenting models with proxies for hidden confounders overcomes the challenges that hidden confounding shifts impose. Together, these results deliver both a new theoretical foundation and concrete, actionable guidance for designing robust OOD generalization algorithms and principled covariate selection strategies.

1 Introduction

Generalization—the ability to draw reliable conclusions about unseen data based on observed data—is central to numerous scientific fields. In medicine and the social sciences, it is embodied as *external validity*, ensuring findings from one population are applicable to a different population [13]; in ecology, it supports *space-for-time substitutions*, where spatial variation is used as a proxy for temporal change to infer long-term ecological patterns [60]; and in engineering, it drives *robust control*, where models must maintain performance in the presence of unmodeled disturbances [35]. In recent years, machine learning has become a powerful tool for learning generalizable models [100, 88, 49].

To generalize well, machine learning models must be robust to distribution shifts between training and test data. Consider a model trained to predict *food stamp reciprocity* from *household attributes* in one state: it must adapt and perform well when deployed in another. A model achieves *out-of-distribution (OOD) generalization* precisely when it maintains its performance on both in-distribution (ID) data (from which training data are sampled) and out-of-distribution (OOD) test data. Over the last decade, entire families of methods—domain generalization [56, 31, 53, 99, 6, 38, 74], domain adaptation [98, 82, 51, 90, 91, 81], and robust learning [42, 69, 94]—have delivered provable OOD generalization, but only under specific assumptions. Once model selection is done carefully [30, 38, 87], models based on standard *empirical risk minimization (ERM)* [86] match or beat these dedicated methods across a broad range of real-world applications [30, 24, 50, 57, 67, 87]. A recent empirical study [57] on 16 real-world tabular datasets goes further, showing that using all available covariates when predicting the outcome—regardless of whether they directly affect it—improves OOD generalization performance. These findings directly contradict prevailing assumptions in the field and demand a deeper investigation into the mechanisms that actually drive OOD generalization.

Distribution shifts arise whenever data originate from different environments [71]. *Food stamp reciprocity*, for instance, differs across states because each state operates under different eligibility rules, producing distinct, environment-specific distributions. Within such data, certain statistical relationships between covariates and the outcome remain consistent across all environments; these

are the *invariant relationships* [6]. Identifying and learning them yields models that generalize out of distribution [6, 59, 31, 66, 56, 98, 63]. A particularly strong form of invariance is *causal invariance*, where the causal relationships in the data stay invariant across environments [71, 59, 6]. Crucially, though, real-world settings rarely observe every covariate relevant to the outcome—data collection is limited, privacy is constrained, and measurements are noisy [14, 52, 20, 48]. This missingness breaks the very invariant relationships that generalization depends on. The problem becomes acute when the unobserved variables are confounders that influence both the observed covariates and the outcome. Such hidden confounders are pervasive in practice, and shifts in their distributions define distinct environments. Ignoring these shifts does more than degrade generalization performance [40, 4, 85, 62]—it drives models to learn outright incorrect relationships between observed covariates and the outcome.

Despite its importance, the impact of hidden confounding shifts on OOD generalization remains largely unexplored [4, 40, 85, 62]. This work closes that gap, providing both the theoretical insights and the empirical evidence that explain the recent findings in the literature [30, 24, 50, 57, 67, 87]. Our contributions are as follows:

- We establish, from a causal perspective, why confounding shift is central to OOD generalization (§ 3), and explain precisely why adding informative, non-causal covariates improves performance.
- We prove that maximizing predictive information between model predictions and true outcomes requires explicitly learning environment-specific relationships. Invariant representations alone, though necessary, are therefore insufficient (§ 4.1, § 4.2).
- We show that variables informative of either the outcome or the hidden confounders directly increase the predictive information between model predictions and true outcomes, establishing that principled covariate selection is essential under hidden confounding shift (§ 4.3).
- Our experiments on real-world and synthetic datasets demonstrate that (i) hidden confounding is prevalent in real-world tabular benchmark data, (ii) learning environment-specific relationships is positively correlated with ID and OOD test accuracy, and (iii) incorporating informative, non-causal covariates improves generalization (§ 5).

In short, our work explains the surprising effectiveness of ERM under real-world distribution shifts and pinpoints hidden confounding as the subtle yet unavoidable driver of OOD generalization—making clear that future methods must confront it head-on.

2 Related work

Out-of-distribution (OOD) generalization spans several facets, including domain generalization [56, 31, 53, 99, 6, 38], domain adaptation [98, 82, 51, 90, 91, 81], robust learning [42, 69, 94], federated learning [54, 36, 44, 97, 45], and OOD detection [41, 32, 92, 84, 49]. Domain adaptation assumes access to unlabeled test-set data, whereas domain generalization relies on environment labels during training. Federated learning adopts a collaborative framework that tackles constraints such as communication efficiency and privacy when data originate from multiple environments, and OOD detection identifies samples that differ significantly from the training distribution. Most of these methods share a single goal: learning invariant relationships. Recent work makes clear, however, that invariance alone does not suffice—additional inductive biases are required for strong generalization [46, 72, 93]. Ye et al. [93] show that feature invariance is necessary but not sufficient and highlight the role of feature *informativeness*. We build on this, studying invariance and informativeness together as the two levers that govern OOD generalization.

Proxy variable adjustment: When confounding variables are observed during training but unobserved at test time, Landeiro and Culotta [40] adjust for confounding shifts to improve classifier performance. Building on proxy-based adjustment for causal effect estimation [55, 39], recent domain adaptation methods recast unknown distribution shifts as a causal effect identification problem [4, 85], and OOD generalization under hidden confounding shift has been studied under an overlapping confounder support assumption [62]. Where these approaches apply proxy variables, we go further and *explain precisely how* proxy variables enhance OOD generalization.

All variable models vs causal models. Nastl and Hardt [57] introduce a benchmark study that sorts covariates into four groups: causal (conservatively chosen), arguably causal, anti-causal, and other spurious covariates. Across 16 benchmark datasets, models using all covariates Pareto-dominate those restricted to causal or arguably causal subsets on both ID and OOD data. Theory

explaining this result has been missing; we supply it, presenting scenarios and arguments that account for their experimental findings. For linear causal models, anchor regression [68] balances two estimation paradigms—models that include all observed covariates and models that use only causal covariates—and we explain the impact of adding non-causal covariates under a hidden confounding shift. Eastwood et al. [21] show that unstable covariates boost performance when they carry information about the label, provided they are conditionally independent of the stable covariates given the label, and propose adjusting the shift by inspecting the unlabeled test domain. Tellingly, on a real-world medical dataset not constructed for this problem [8], ERM remains competitive with their method, exactly as Nastl and Hardt [57] report. This is a manifestation of a broader fact: under well-specified covariate shifts, maximum likelihood estimation (MLE) is minimax optimal for OOD generalization [26]. Real-world settings, however, are almost never well-specified—precisely because of hidden confounding shift, the central focus of this work.

3 Manifestations of hidden confounding shift

In this section, we lay out the background on hidden confounding shift and demonstrate why it must be addressed, using an example that turns on causal effect identification of the observed covariates on the outcome.

Types of distribution shifts. For covariates $\mathbf{X}$ and target Y , two environments e and e' can exhibit any of the distribution shifts in Table 1. Each of these shifts traces back to a single source: a shift in the distribution $\mathbb{P}(U)$ of an unobserved covariate U that causes $\mathbf{X}$, Y , or both (see Figure 1). A shift in $\mathbb{P}(U)$ propagates into the observed distributions over $\mathbf{X}$ and Y . To see this, write $\mathbb{P}(\mathbf{X}) = \sum_U \mathbb{P}(U) \mathbb{P}(\mathbf{X} | U)$: whenever $\mathbb{P}^e(U) \neq \mathbb{P}^{e'}(U)$ and $U \rightarrow \mathbf{X}$, we necessarily observe $\mathbb{P}^e(\mathbf{X}) \neq \mathbb{P}^{e'}(\mathbf{X})$. The same argument carries over verbatim to $\mathbb{P}(Y)$, $\mathbb{P}(\mathbf{X} | Y)$, $\mathbb{P}(Y | \mathbf{X})$, and $\mathbb{P}(\mathbf{X}, Y)$.

Existing OOD generalization methods lean on specific causal relationships among U , $\mathbf{X}$, Y that guarantee particular invariances. When $U \rightarrow Y \rightarrow \mathbf{X}$ (Figure 1 (a)), since $\mathbb{P}(\mathbf{X}, Y) = \mathbb{P}(Y) \mathbb{P}(\mathbf{X} | Y)$, the observed shift $\mathbb{P}^e(\mathbf{X}, Y) \neq \mathbb{P}^{e'}(\mathbf{X}, Y)$ is pure label shift, i.e., $\mathbb{P}^e(Y) \neq \mathbb{P}^{e'}(Y)$, while the conditional covariate distribution stays invariant, $\mathbb{P}^e(\mathbf{X} | Y) = \mathbb{P}^{e'}(\mathbf{X} | Y)$, because $\mathbb{P}(\mathbf{X} | Y, U) = \mathbb{P}(\mathbf{X} | Y)$ [89, 83, 25, 5, 47, 96, 18, 79]. Symmetrically, when $U \rightarrow \mathbf{X} \rightarrow Y$ (Figure 1 (b)), the shift is pure covariate shift, $\mathbb{P}^e(\mathbf{X}) \neq \mathbb{P}^{e'}(\mathbf{X})$, while $\mathbb{P}(Y | \mathbf{X})$ stays invariant, $\mathbb{P}^e(Y | \mathbf{X}) = \mathbb{P}^{e'}(Y | \mathbf{X})$ [70, 16, 80, 11, 28]. These clean invariances collapse in the realistic case, because U typically causes *both* $\mathbf{X}$ and Y (Figure 1 (c)) [50, 4, 40, 85, 62, 65, 64]. Here a single shift in $\mathbb{P}(U)$ triggers label shift, covariate shift, conditional covariate shift, and concept shift all at once. The only invariance that survives is $\mathbb{P}^e(\mathbf{X}, Y | U) = \mathbb{P}^{e'}(\mathbf{X}, Y | U)$, the *stable confounding shift* [85, 4]—and because U is unobserved, even this invariance never appears in the observed data.

Information-theoretic measures: Following Federici et al. [22], we quantify distribution shifts with mutual information. We use $I(\mathbf{X}; E)$, $I(Y; E)$, $I(\mathbf{X}; E | Y)$, $I(Y; E | \mathbf{X})$, $I(\mathbf{X}, Y; E)$, where E is the environment variable, to measure the shifts in $\mathbb{P}(\mathbf{X})$, $\mathbb{P}(Y)$, $\mathbb{P}(\mathbf{X} | Y)$, $\mathbb{P}(Y | \mathbf{X})$, $\mathbb{P}(\mathbf{X}, Y)$, respectively, across environments. Existing OOD generalization methods assume either $I(\mathbf{X}; E | Y) = 0$ (the label shift assumption) or $I(Y; E | \mathbf{X}) = 0$ (the covariate shift assumption). Under hidden confounding shift, both quantities are generally nonzero, so both assumptions fail.

Challenges with hidden confounding shift: The experiment in Figure 2 makes the challenge concrete. The outcome Y depends on X and a hidden confounder U , and environment-specific shifts in the mean of U , denoted μ_e , induce distinct environments. A linear regression model trained solely on the observed covariate X infers the *wrong* relationship between X and Y (first subplot): this is Simpson’s paradox [73]— Y increases with X within each environment, yet the model captures

Table 1: The distribution shifts a hidden confounder shift can induce.

Type of Shift	Mathematical Expression
Label	$\mathbb{P}^e(Y) \neq \mathbb{P}^{e'}(Y)$
Covariate	$\mathbb{P}^e(\mathbf{X}) \neq \mathbb{P}^{e'}(\mathbf{X})$
Conditional Covariate	$\mathbb{P}^e(\mathbf{X} Y) \neq \mathbb{P}^{e'}(\mathbf{X} Y)$
Concept	$\mathbb{P}^e(Y \mathbf{X}) \neq \mathbb{P}^{e'}(Y \mathbf{X})$
Dataset	$\mathbb{P}^e(\mathbf{X}, Y) \neq \mathbb{P}^{e'}(\mathbf{X}, Y)$

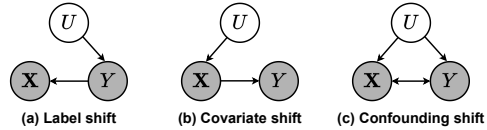


Figure 1: Causal graphs underlying distribution shifts.

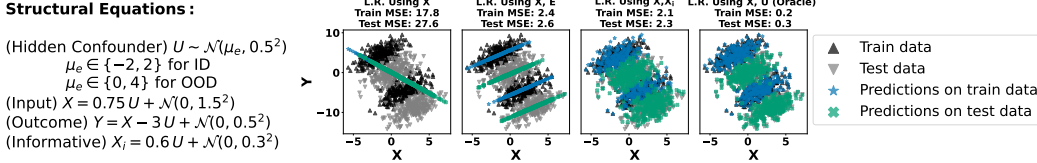


Figure 2: Four linear regression (L.R.) models in an OOD setting with hidden confounding shifts and minimal environment overlap (distant μ_e). (i) Training solely on X learns the wrong relationship with Y —Simpson’s paradox. (ii) Environment-specific summary statistics of X , denoted E , recover the correct relationship, but their representational power is limited. (iii) Adding an informative covariate X_i for U improves OOD generalization. (iv) The oracle model is trained on X and U .

the opposite trend. Methods purpose-built for OOD generalization—IRM [6], VREX [38], and Group DRO [69]—fail in exactly the same way. We prove in § 4 that the optimal generalization strategy is to learn environment-specific relationships, which recovers the correct X – Y relationship. The second subplot confirms this: environment-specific summary statistics of X (mean, standard deviation, quantiles) uncover the true relationship, mirroring the *backdoor adjustment* criterion in causal effect estimation [58], with environment-specific information acting as a proxy for the confounder. Yet observed covariates alone carry limited representational capacity, so this correction remains partial. As we show in § 4.3, adding informative covariates that serve as proxies for the unobserved confounder pushes generalization further still (third subplot), approaching the oracle model trained on X and U , which attains the best-possible mean squared error.

4 Impact of hidden confounding shifts on OOD generalization

In this section, we establish theoretically how hidden confounding shifts impact every aspect of OOD generalization. Let $\hat{Y} = (f \circ \phi)(\mathbf{X})$ be the predicted label for an input $\mathbf{X}$, where f is a classifier and ϕ is a feature extractor or transformation function. $H(X) = -\mathbb{E}_X[\log(\mathbb{P}(X))]$ denotes entropy and $I(X; Y) = \mathbb{E}_{X,Y}[\log \frac{\mathbb{P}(X,Y)}{\mathbb{P}(X)\mathbb{P}(Y)}]$ denotes mutual information. The risk of the predictor $(f \circ \phi)$ in an environment e is defined as: $\mathcal{R}^e(f \circ \phi) = \mathbb{E}_{(\mathbf{X}, Y) \sim \mathbb{P}_{\mathbf{X}, Y}^e}[\ell((f \circ \phi)(\mathbf{X}), Y)]$ where $\ell(\cdot, \cdot)$ is a loss function. The goal in OOD generalization is to learn a predictor $(f \circ \phi)$ that performs well on both ID and OOD data. To this end, various objectives have been considered in the literature. Robust optimization-based methods [7, 9, 19, 75, 69] aim to minimize the worst risk across all training environments: $\mathcal{R}^{\text{Rob}}(f \circ \phi) = \max_{e \in \mathcal{E}_{tr}} \mathcal{R}^e(f \circ \phi)$ where $\mathcal{E}_{tr}$ denotes the set of training environments. The variance risk extrapolation (VREX) [38] aims to minimize $\mathcal{R}^{\text{VREX}}(f \circ \phi) = \sum_{e \in \mathcal{E}_{tr}} \mathcal{R}^e(f \circ \phi) + \beta \times \text{Variance}(\{\mathcal{R}^1(f \circ \phi), \dots, \mathcal{R}^{|\mathcal{E}_{tr}|}(f \circ \phi)\})$ where β is a hyperparameter. Invariant risk minimization (IRM) [6] aims to minimize $\sum_{e \in \mathcal{E}_{tr}} \mathcal{R}^e(f \circ \phi)$ with the constraint that f is a simultaneously optimal classifier across all environments. Empirical risk minimization (ERM) based methods simply pool data from all training environments and minimize the empirical risk on the pooled data [6, 38]. That is, ERM minimizes $\mathcal{R}^{\text{ERM}}(f \circ \phi) = \sum_{e \in \mathcal{E}_{tr}} \sum_{(\mathbf{X}, Y) \in \mathcal{D}^e} \ell((f \circ \phi)(\mathbf{X}), Y)$ where $\mathcal{D}^e$ denotes observed data in an environment e .

To pin down how hidden confounding shift impacts traditional objective functions, we analyze the maximization of predictive information $I(Y; \hat{Y})$, where $\hat{Y} = (f \circ \phi)(\mathbf{X})$. Since $I(Y; \hat{Y})$ measures exactly how informative the prediction $\hat{Y}$ is about the true label Y , it is the natural objective for a wide range of machine learning tasks. We begin by defining the two properties of $\phi(\mathbf{X})$ that govern OOD generalization [93]: *informativeness* and *invariance*.

Definition 4.1 (Informativeness and Conditional Informativeness). *The informativeness of features $\phi(\mathbf{X})$ for predicting Y is defined as $\text{INF}(\phi(\mathbf{X}), Y) = I(\phi(\mathbf{X}); Y)$. The conditional informativeness of features $\phi(\mathbf{X})$ for predicting Y conditioned on environment variable E is defined as $\text{CINF}(\phi(\mathbf{X}), Y, E) = I(\phi(\mathbf{X}); Y|E)$.*

Conditional informativeness measures the information $\phi(\mathbf{X})$ carries about Y within each environment. Minimizing CINF discards information outright, and maximizing it can be undesirable in settings such as algorithmic fairness, where $\phi(\mathbf{X})$ may exploit sensitive or biased within-environment information and produce unfair predictions. Minimizing $I(\phi(\mathbf{X}); E)$ removes such biased representations, but at a cost to predictive performance, as we show in § 4.2. When no sensitive information is tied to E , maximizing CINF improves generalization—a claim we substantiate in § 4.1.

Definition 4.2 (Variation and Invariance). *For a given label Y , the variation in the features $\phi(\mathbf{X})$ across environments E is defined as $\text{VAR}(\phi(\mathbf{X}), Y, E) = I(\phi(\mathbf{X}); E|Y)$. For a given label Y , the invariance of the features $\phi(\mathbf{X})$ across environments E is defined as the negative of the variation i.e., $\text{INV}(\phi(\mathbf{X}), Y, E) = -\text{VAR}(\phi(\mathbf{X}), Y, E) = -I(\phi(\mathbf{X}); E|Y)$*

Invariance quantifies how consistent the representation $\phi(\mathbf{X})$ is across environments for a given label Y . Minimizing invariance invites overfitting by retaining environment-specific information in $\phi(\mathbf{X})$, whereas maximizing it strips out environment-specific dependencies and drives invariant learning. Extending the measures from § 3, we use $I(\phi(\mathbf{X}); E)$ and $I(Y; E | \phi(\mathbf{X}))$ to quantify *feature shift* and *concept shift*, respectively. Depending on context, *concept shift* denotes either $I(Y; E | \mathbf{X})$ or $I(Y; E | \phi(\mathbf{X}))$.

4.1 A general decomposition of predictive information

We begin with a few causal-graph preliminaries. A causal graph $\mathcal{G}$ consists of nodes representing random variables, and directed edges indicating direct causal influences between nodes. A *path* between two nodes X_i and X_j is a sequence of unique nodes connected by edges. A *directed path* from X_i to X_j with $i < j$ is one where all arrows point toward X_j , i.e., $X_i \rightarrow X_{i+1} \rightarrow \dots \rightarrow X_{j-1} \rightarrow X_j$. In such a directed path, X_i is called the *parent* of X_{i+1} , X_{i+1} is the *child* of X_i , X_i an *ancestor* of X_j , and X_j is a *descendant* of X_i . Paths decompose into three fundamental structures: a *chain* $X_i \rightarrow X_j \rightarrow X_k$, a *fork* $X_i \leftarrow X_j \rightarrow X_k$, and a *collider* $X_i \rightarrow X_j \leftarrow X_k$. In both chains and forks, X_i and X_k are marginally dependent yet become conditionally independent upon conditioning on the intermediate node X_j , i.e. $X_i \not\perp\!\!\!\perp X_k$ and $X_i \perp\!\!\!\perp X_k | X_j$. In a collider, X_i and X_k are marginally independent but become conditionally dependent when conditioning on X_j or any of its descendants, i.e. $X_i \perp\!\!\!\perp X_k$ and $X_i \not\perp\!\!\!\perp X_k | X_j$. A path between X_i and X_k is said to be *blocked* by a conditioning set $\mathcal{S}$ if and only if either (i) the path contains a chain or fork whose middle node lies in $\mathcal{S}$, or (ii) it contains a collider such that neither the collider nor any of its descendants belongs to $\mathcal{S}$. If all paths from X_i to X_k are blocked by $\mathcal{S}$, then $X_i \perp\!\!\!\perp X_k | \mathcal{S}$. A path is open if it is not blocked.

Now consider the predictive information $I(Y; \hat{Y})$, where the predictions $\hat{Y} = (f \circ \phi)(\mathbf{X})$ rest on a learned representation $\phi(\mathbf{X})$. We model the causal relationships among $\mathbf{X}, Y, U, \phi(\mathbf{X}), \hat{Y}$ and E as in Figure 3, where U is the hidden confounding variable. Given either $\mathbf{X}$ or $\phi(\mathbf{X})$, $\hat{Y}$ adds nothing to reasoning about Y : $\hat{Y} \perp\!\!\!\perp Y | \mathbf{X}$ and $\hat{Y} \perp\!\!\!\perp Y | \phi(\mathbf{X})$. The environment variable E captures shifts in $\mathbb{P}(U)$, so that any two environments differ by a shift in $\mathbb{P}(U)$. We first derive a general decomposition of predictive information that makes no assumption about how U influences $\mathbf{X}, Y$; we then give the formal treatment of U 's influence in § 4.2.

Proposition 4.1. *For a covariate vector $\mathbf{X}$, label Y , with causal structure $\mathbf{X} \leftrightarrow Y$, i.e., some covariates cause Y and some covariates are caused by Y , environment variable E , a feature extractor ϕ , and prediction $\hat{Y}$, the predictive information $I(Y; \hat{Y})$ is decomposed as follows:*

$$I(Y; \hat{Y}) = \underbrace{I(\phi(\mathbf{X}); Y|E)}_{\text{Cond. informativ.}} - \underbrace{\frac{I(\phi(\mathbf{X}); E|Y)}{2}}_{\text{Variation}} + \underbrace{\frac{I(Y; E)}{2}}_{\text{Label shift}} + \underbrace{\frac{I(\phi(\mathbf{X}); E)}{2}}_{\text{Feature shift}} - \underbrace{\frac{I(Y; E|\phi(\mathbf{X}))}{2}}_{\text{Concept shift}} - \underbrace{I(\phi(\mathbf{X}); Y|\hat{Y})}_{\text{Residual}} \quad (1)$$

where $I(\phi(\mathbf{X}); Y|\hat{Y})$ is the residual information in $\phi(\mathbf{X})$ for inferring Y that is not captured by the prediction $\hat{Y}$. The decomposition above also holds when $\phi(\mathbf{X})$ is replaced with $\mathbf{X}$.

Proofs are in Appendix § A. This decomposition unifies existing methods: each one explicitly minimizes or maximizes a specific term in Equation (1). IRM [6] forces a fixed classifier f to remain optimal across environments, which is exactly maximizing $-I(\phi(\mathbf{X}); E|Y)$ —any extra environment-specific information in $\phi(\mathbf{X})$ conditional on Y would break the classifier's cross-environment optimality. DANN [23] and the independence criterion for fair classification [22] minimize environment-specific information in $\phi(\mathbf{X})$ by minimizing $I(\phi(\mathbf{X}); E)$, and domain adaptation methods learn features invariant to domains [82, 90]. Minimizing $I(\phi(\mathbf{X}); E)$ serves fair classification but demonstrably degrades predictive performance [98, 33, 22]: Equation (1) shows directly that reducing $I(\phi(\mathbf{X}); E)$ reduces $I(Y; \hat{Y})$. CDAN [51] enforces $\mathbb{P}^e(\phi(\mathbf{X}), Y) = \mathbb{P}^{e'}(\phi(\mathbf{X}), Y)$, making E independent of the pair $(\phi(\mathbf{X}), Y)$; consequently $\mathbb{P}(Y | \phi(\mathbf{X}), E) = \mathbb{P}(Y | \phi(\mathbf{X})) \implies I(Y; E |$

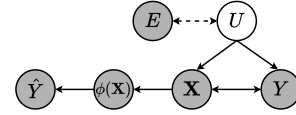


Figure 3: Bi-directed arrow between $\mathbf{X}$ and Y indicate that some covariates of $\mathbf{X}$ can cause Y , and some may be caused by Y .

243 $\phi(\mathbf{X}) = 0$. So even though CDAN never writes $I(Y; E | \phi(X))$ into its loss, its joint-distribution
 244 alignment drives this conditional mutual information to zero. Finally, while $I(Y; E)$ is a constant,
 245 several methods are explicitly robust to shifts in the label distribution [69, 95].

246 Equation (1) makes no assumption about how U influences $\mathbf{X}, Y$, but knowing that structure sharpens
 247 how we should handle each term. If $U \rightarrow Y, U \not\rightarrow \mathbf{X}, Y \rightarrow \mathbf{X}$, then $I(\mathbf{X}; E | Y) = 0$, so
 248 minimizing $I(\phi(\mathbf{X}); E | Y)$ is the principled objective. Symmetrically, if $U \not\rightarrow Y, U \rightarrow \mathbf{X}, \mathbf{X} \rightarrow Y$,
 249 then $I(Y; E | \mathbf{X}) = 0$, which motivates minimizing $I(Y; E | \phi(\mathbf{X}))$. The hard case—and the
 250 realistic one—is $U \rightarrow Y$ and $U \rightarrow \mathbf{X}$, where the terms in Equation (1) interact non-trivially and the
 251 right strategy is far from obvious. We resolve this case in the next section.

252 4.2 Predictive information under hidden confounding shift

253 To determine how a hidden confounder U that causes both $\mathbf{X}$ and Y shapes the predictive information,
 254 we analyze the *two special cases* of the $\mathbf{X}$ – Y relationship: (i) $\mathbf{X} \rightarrow Y$ and (ii) $Y \rightarrow \mathbf{X}$. When
 255 $\mathbf{X} \rightarrow Y$, the inequalities in (2) hold because conditioning on Y opens the path $\phi(\mathbf{X}) \leftarrow \mathbf{X} \rightarrow Y \leftarrow$
 256 $U \leftarrow E$ from $\phi(\mathbf{X})$ to E at the collider node Y , adding information flow from $\phi(\mathbf{X})$ to E via Y ;
 257 conditioning on $\phi(\mathbf{X})$, in contrast, partially blocks the flow from Y to E at $\mathbf{X}$, as long as $\phi(\mathbf{X})$
 258 encodes some information about $\mathbf{X}$.

$$(i) \overbrace{I(\phi(\mathbf{X}); E | Y)}^{\text{Variation}} \geq \overbrace{I(\phi(\mathbf{X}); E)}^{\text{Feature shift}} \quad (ii) \overbrace{I(Y; E)}^{\text{Label shift}} \geq \overbrace{I(Y; E | \phi(\mathbf{X}))}^{\text{Concept shift}} \quad (2)$$

259 Symmetrically, when $Y \rightarrow \mathbf{X}$, the inequalities in (3) hold because conditioning on Y blocks the flow
 260 from $\phi(\mathbf{X})$ to E at Y , while conditioning on $\phi(\mathbf{X})$ opens the path $Y \rightarrow \mathbf{X} \leftarrow U \leftarrow E$ from Y to E
 261 at $\mathbf{X}$ —because $\phi(\mathbf{X})$ is a child of $\mathbf{X}$, and conditioning on the child of a collider opens a path through
 262 that collider [58] (recall the preliminaries in § 4.1).

$$(i) \overbrace{I(\phi(\mathbf{X}); E)}^{\text{Feature shift}} \geq \overbrace{I(\phi(\mathbf{X}); E | Y)}^{\text{Variation}} \quad (ii) \overbrace{I(Y; E | \phi(\mathbf{X}))}^{\text{Concept shift}} \geq \overbrace{I(Y; E)}^{\text{Label shift}} \quad (3)$$

263 The inequalities in (2) and (3) let us collapse the predictive information decomposition to its essential
 264 form.

265 **Proposition 4.2.** *For a covariate vector $\mathbf{X}$, label Y , an environment variable E , a feature extractor*
 266 *ϕ , the prediction $\hat{Y}$, and an unobserved confounding variable U that cause both $\mathbf{X}$ and Y , and if*
 267 *either (i) $\mathbf{X} \rightarrow Y$ or (ii) $Y \rightarrow \mathbf{X}$, the predictive information $I(Y; \hat{Y})$ can be decomposed as follows:*

$$I(Y; \hat{Y}) = \overbrace{I(\phi(\mathbf{X}); Y | E)}^{\text{Cond. informativeness}} - \overbrace{I(\phi(\mathbf{X}); Y | \hat{Y})}^{\text{Residual}} \quad (4)$$

268 Proposition 4.2 handles the two cases $\mathbf{X} \rightarrow Y$ and $Y \rightarrow \mathbf{X}$ separately, but $\mathbf{X} \rightarrow Y$ dominates
 269 real-world tabular prediction tasks: 11 of the 16 datasets in recent benchmark studies [57, 50, 24]
 270 follow $\mathbf{X} \rightarrow Y$, and in the remaining 5, the covariates caused by Y are far fewer than those that cause
 271 Y . We state the key takeaway from Proposition 4.2 below.

Compared to the decomposition in (1), the decomposition in (4) (which is obtained for the special cases: (i) $U \rightarrow \mathbf{X}, U \rightarrow Y, \mathbf{X} \rightarrow Y$, (ii) $U \rightarrow \mathbf{X}, U \rightarrow Y, Y \rightarrow \mathbf{X}$) is free from the terms: *variation, label shift, feature shift, and concept shift*. That is, under hidden confounding shift, when either $\mathbf{X} \rightarrow Y$ or $Y \rightarrow \mathbf{X}$, the predictive information can be maximized by maximizing the difference: *conditional informativeness* – *residual*.

272

273 **Maximizing conditional informativeness:** In recent work, Prashant et al. [62] proposed a mixture
 274 of experts (MoE) model for OOD generalization under hidden confounding shift, where each expert
 275 corresponds to a specific hidden confounder assignment and maximizes performance within the
 276 environment for that confounder value. Learning such MoE models is precisely maximizing the
 277 conditional informativeness $I(\phi(\mathbf{X}); Y | E)$, and Equation (4) supplies the theoretical justification
 278 for them. It also does more: it exposes the need for methods that operate under far more general
 279 confounding shift settings, since Prashant et al. [62] assume overlapping confounding support,
 280 $\text{supp}(\mathbb{P}(U | \mathcal{E}_{te})) \subseteq \text{supp}(\mathbb{P}(U | \mathcal{E}_{tr}))$, and require discrete-valued proxy variables.

The results in Figure 2 show that, even without confounding overlap and with a continuous-valued proxy, a linear regression model recovers the correct causal relationship between observed covariates and the target across environments once given environment-specific information. This concrete example makes the case for OOD generalization methods that stay effective when confounding support is limited and the confounder is continuous-valued.

Minimizing residual: The residual term $I(\phi(\mathbf{X}); Y | \hat{Y})$ measures how much additional information $\phi(\mathbf{X})$ carries about Y beyond what the prediction $\hat{Y}$ already captures, and decomposes as $I(\phi(\mathbf{X}); Y | \hat{Y}) = H(Y | \hat{Y}) - H(Y | \phi(\mathbf{X}), \hat{Y})$. The conditional entropy $H(Y | \hat{Y})$ ties directly to the expected cross-entropy loss: $\mathbb{E}_{Y, \hat{Y}}[\ell_{\text{CE}}(Y, \hat{Y})] = H(Y | \hat{Y}) + \text{calibration error}$ [12, 10, 29], where $H(Y | \hat{Y})$ is the *refinement error* capturing the model’s ability to distinguish between classes. Minimizing the residual therefore admits two complementary strategies. First, minimize $I(\phi(\mathbf{X}); Y | \hat{Y})$ directly by making $\phi(\mathbf{X})$ conditionally independent of Y given $\hat{Y}$, i.e., $Y \perp\!\!\!\perp \phi(\mathbf{X}) | \hat{Y}$. Second, minimize the refinement error $H(Y | \hat{Y})$ —for example, through training techniques that drive down refinement error for accurate predictions before addressing calibration error [10].

4.3 Impact of additional informative variables

In § 4.2, we established the optimal generalization strategies when the confounder is unobserved. In practice, we often have access to proxies for the hidden confounder [4, 85, 62] that can stand in for it. We now quantify exactly what adding such non-causal but informative covariates does to the predictive information.

Definition 4.3 (Informative Covariates). *A set of covariates $\mathbf{X}_I$ that are not causally related to Y i.e., neither ancestors nor descendants of Y are in $\mathbf{X}_I$, are said to be informative to Y if $\mathbf{X}_I$ and Y are not independent of each other given other causally related covariates $\mathbf{X}$ and E i.e., $Y \not\perp\!\!\!\perp \mathbf{X}_I | \mathbf{X}, E$.*

Since $U \rightarrow Y$, any covariate informative of U is informative of Y , and vice versa. The decomposition in (1) makes clear that maximizing $I(Y; \hat{Y})$ requires minimizing the concept shift $I(Y; E | \phi(\mathbf{X}))$. Liu et al. [50] report empirically that collecting additional covariates $\mathbf{X}_I$ that make $\mathbb{P}(Y | \mathbf{X}, \mathbf{X}_I)$ more stable across environments—i.e., reduce concept shift—improves OOD test accuracy. We now prove that using more informative variables maximizes predictive information by moving specific terms in (1) in their favorable directions.

Proposition 4.3. *If $\phi_1(\cdot), \phi_2(\cdot)$ are invertible functions and if $\mathbf{X}_I$ are informative variables such that $Y \not\perp\!\!\!\perp \mathbf{X}_I | \mathbf{X}, E$, then we have the following inequalities:*

$$\begin{aligned} (i) \ I(\phi_2(\{\mathbf{X} \cup \mathbf{X}_I\}); Y | E) &> I(\phi_1(\mathbf{X}); Y | E) & (ii) \ I(\phi_2(\{\mathbf{X} \cup \mathbf{X}_I\}); E) &> I(\phi_1(\mathbf{X}); E) \\ (iii) \ I(Y; E | \phi_2(\{\mathbf{X} \cup \mathbf{X}_I\})) &< I(Y; E | \phi_1(\mathbf{X})) & (iv) \ I(\phi_2(\{\mathbf{X} \cup \mathbf{X}_I\}); E | Y) &> I(\phi_1(\mathbf{X}); E | Y) \end{aligned} \quad (5)$$

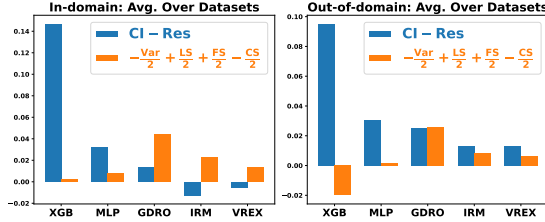
Proposition 4.3 establishes that adding informative covariates increases conditional informativeness (5.i) and feature shift (5.ii) while reducing concept shift (5.iii). The one countervailing effect is that it also amplifies variation (5.iv), which calls for dedicated strategies to control that variation.

5 Experimental results

We run experiments on both real-world and synthetic datasets to test three claims: (i) hidden confounding shift is present in real-world data, (ii) the components of the decomposition in (1) govern OOD generalization under hidden confounding shifts, and (iii) informative covariates improve generalization. We use eight real-world tabular benchmark datasets—*Food stamps*, *Readmission*, *Income*, *Public coverage*, *Unemployment*, *Diabetes*, *Hypertension*, and *ASSISTments*—with the datasets and domain splits taken from the *TableShift* benchmark [24]. We measure performance with ID test and OOD test accuracies, and evaluate two ERM-based methods (XGBoost (XGB) [15] and a multi-layer perceptron (MLP)), two domain generalization methods (IRM [6], VREX [38]), and one robust learning method (Group DRO (GDRO) [69]). We report averages over 5 random hyperparameter choices (Appendix B); random seeds have a statistically insignificant effect on these datasets [24]. Full experimental details are in Appendix § B, and results with hyperparameter tuning are in Appendix C.

Table 2: Quantifying distribution shifts. Mean $\pm$ standard deviation over 10 random subsets of 40,000 samples. One-sample t-tests against a mean of zero ($\alpha = 0.05$) confirm that every measure is significantly nonzero (p-value ≈ 0)—hidden confounding shift is unambiguously present.

Dataset	Conditional Covariate Shift $I(\mathbf{X}; E Y)$	Label shift $I(Y; E)$	Covariate Shift $I(\mathbf{X}; E)$	Concept shift $I(Y; E \mathbf{X})$
Readmission	0.107 ± 0.002	0.068 ± 0.002	0.097 ± 0.002	2.032 ± 0.000
Food stamps	0.126 ± 0.004	0.030 ± 0.003	0.108 ± 0.001	2.118 ± 0.001
Income	0.168 ± 0.002	0.075 ± 0.003	0.147 ± 0.001	2.059 ± 0.002
Public coverage	0.231 ± 0.002	0.412 ± 0.006	0.222 ± 0.002	1.945 ± 0.001
Unemployment	0.117 ± 0.001	0.019 ± 0.002	0.114 ± 0.002	2.010 ± 0.003
Diabetes	0.032 ± 0.002	0.048 ± 0.001	0.022 ± 0.002	2.132 ± 0.001
Hypertension	0.090 ± 0.002	0.183 ± 0.003	0.037 ± 0.001	1.883 ± 0.004
ASSISTments	0.293 ± 0.002	0.260 ± 0.001	0.306 ± 0.002	0.367 ± 0.004



Method	Avg. ID Test Acc.	Avg. OOD Test Acc.
XGB	85.22	77.00
MLP	83.64	73.29
GDRO	78.40	72.28
IRM	71.17	64.89
VREX	59.74	63.06

Figure 4: The difference *conditional informativeness* – *residual* tracks the average ID and OOD test accuracy over the eight datasets (table, right): higher values of the difference correspond to higher accuracy, exactly as Proposition 4.2 predicts.

Hidden confounding shift: Hidden confounding shifts induce observable shifts in $\mathbb{P}(\mathbf{X})$, $\mathbb{P}(\mathbf{X} | Y)$, $\mathbb{P}(Y)$, and $\mathbb{P}(Y | \mathbf{X})$. Table 2 shows that label shift, covariate shift, conditional covariate shift, and concept shift are all present—and statistically significant—in the real-world datasets, confirming that hidden confounding shifts are at work. To complement this quantitative evidence, we query GPT-4o [1] to list plausible hidden confounders for several benchmark datasets [24, 57] (Appendix § E). In the *food stamps* dataset, for example, unmeasured factors such as state-specific *economic policies* plausibly drive both *household income* and *food stamp reciprocity*.

Conditional informativeness vs. accuracy: Proposition 4.2 establishes that, under hidden confounding shift, maximizing predictive information reduces to maximizing *conditional informativeness* – *residual*. Figure 4 reports this quantity on the first six datasets in Table 2 (those with more than one training environment [57, 24]); dataset-wise results are in Appendix C. The prediction holds: *conditional informativeness* – *residual* is positively correlated with both ID and OOD accuracy, and XGB—the best-performing model—attains the highest value of this difference. The dataset-wise results in Appendix C further confirm the theory: the orange bars sit near zero, exactly as Proposition 4.2 predicts, because the sum $-\text{variation} / 2 + \text{label shift} / 2 + \text{feature shift} / 2 - \text{concept shift} / 2$ vanishes under hidden confounding shift. The relative ID/OOD ranking of the methods in Figure 1 matches previously reported results [57, 24].

Informative covariates vs. accuracy: We now show that including informative covariates yields better OOD generalization. We adopt the covariate partitioning of [57], which groups all available covariates into three nested subsets, *causal covariates* $\subseteq$ *arguably causal covariates* $\subseteq$ *all covariates* (Appendix § B), and measure how the terms in (1) move as we enlarge the covariate set. To do this, we introduce the *sign consistency metric*. Moving from a smaller to a larger covariate set, for any term in (1) with a positive coefficient (e.g., conditional informativeness) we count how often its value rises when accuracy improves; for negatively weighted terms (e.g., concept shift) we count how often its value falls when accuracy improves. The metric is the fraction of observations where a term moves in its *beneficial* direction, capturing how reliably it drives better generalization (formal definition in Appendix § B). Table 3 shows that both conditional informativeness and concept shift are highly sign-consistent with accuracy gains. Reducing concept shift is necessary, but the decisive advantage of XGB is that it further raises conditional informativeness beyond what the other methods achieve, translating into its additional performance gains. Dataset-specific results are in Appendix § C.

Table 3: Every method attains high sign consistency for concept shift (CS). The decisive term for generalization, however, is conditional informativeness (CI), per Proposition 4.2—and XGB clearly excels at it. Res: residual, Var: variation, FS: feature shift, C: causal, AC: arguably causal, A: all.

Method	Sign Consistency Metric ($\uparrow$)					ID-Test Accuracy ($\uparrow$)			OOD-Test Accuracy ($\uparrow$)		
	CI	Var	FS	CS	Res	C	AC	A	C	AC	A
XGB	0.89	0.61	0.36	0.89	0.17	81.24	84.84	85.22	69.92	76.72	77.00
MLP	0.75	0.64	0.28	0.89	0.33	80.51	81.93	83.64	69.70	73.84	73.29
GDRO	0.69	0.67	0.19	0.94	0.36	80.92	83.76	78.40	69.61	73.49	72.28
IRM	0.78	0.61	0.25	0.92	0.25	67.91	68.22	71.17	61.21	62.96	64.89
VREX	0.64	0.61	0.25	0.89	0.44	54.39	62.32	59.74	56.62	63.53	63.06

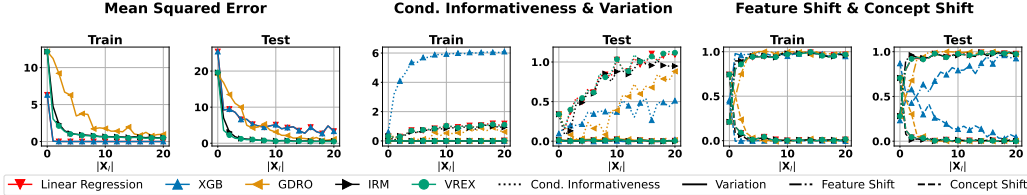


Figure 5: Adding proxy variables X_I of U that are informative of Y reduces MSE, raises conditional informativeness and feature shift, and lowers concept shift.

In real-world datasets, one can rarely verify whether *all available covariates* truly include *every relevant or sufficiently informative* variable. We therefore test our hypothesis directly on synthetic data with a known causal structure: $U \rightarrow X$, $U \rightarrow Y$, $X \rightarrow Y$, and $U \rightarrow X_I$, where X_I are informative covariates caused by the hidden confounder U . Including X_I improves conditional informativeness and feature shift while reducing concept shift, exactly as predicted (Figure 5); the variation component stays at zero throughout, as the simplicity of the setup dictates. The synthetic data generation process and additional results are in Appendix § D.

6 Discussion, limitations, and future work

We decomposed the predictive information between model outputs and true outcomes and, in doing so, exposed the factors that limit OOD generalization (Equation (1)). Hidden confounding shifts—manifested as covariate and outcome distribution changes across environments—are unavoidable, and no finite set of observed covariates or proxies can fully resolve the ambiguity they induce. Because confounding entangles covariate, label, concept, and feature shifts (inequalities in (2), (3)), treating any of them in isolation is doomed to fail. We proved that learning environment-specific input–output mappings beats enforcing a single invariant predictor (Equation (4)). By establishing that hidden confounding is inevitable and must be confronted directly, our work lays the foundation for the next generation of robust OOD methods. The central open challenges follow directly: (i) algorithms that remain effective under limited confounder overlap, (ii) a quantitative account of the cost–accuracy trade-off in acquiring informative covariates, (iii) methods that handle entangled shifts without leaning on untestable proxy assumptions, and (iv) models of the ambiguity from unobserved confounders that will drive genuinely new OOD-robust paradigms.

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652 Appendix

653 A Proofs of propositions

654 **Proposition 4.1.** For a covariate vector $\mathbf{X}$, label Y , with causal structure $\mathbf{X} \leftrightarrow Y$, i.e., some
 655 covariates cause Y and some covariates are caused by Y , environment variable E , a feature
 656 extractor ϕ , and prediction $\hat{Y}$, the predictive information $I(Y; \hat{Y})$ is decomposed as follows:

$$I(Y; \hat{Y}) = \overbrace{I(\phi(\mathbf{X}); Y|E)}^{\text{Cond. informativ.}} - \overbrace{\frac{I(\phi(\mathbf{X}); E|Y)}{2}}^{\text{Variation}} + \overbrace{\frac{I(Y; E)}{2}}^{\text{Label shift}} + \overbrace{\frac{I(\phi(\mathbf{X}); E)}{2}}^{\text{Feature shift}} - \overbrace{\frac{I(Y; E|\phi(\mathbf{X}))}{2}}^{\text{Concept shift}} - \overbrace{I(\phi(\mathbf{X}); Y|\hat{Y})}^{\text{Residual}} \quad (1)$$

657 where $I(\phi(\mathbf{X}); Y|\hat{Y})$ is the residual information in $\phi(\mathbf{X})$ for inferring Y that is not captured by the
 658 prediction $\hat{Y}$. The decomposition above also holds when $\phi(\mathbf{X})$ is replaced with $\mathbf{X}$.

659 *Proof.* From the causal graph shown in Figure 3 of the main paper, in the causal substructure:
 660 $\hat{Y} \leftarrow \phi(\mathbf{X}) \leftarrow \mathbf{X} \leftrightarrow Y$, $\hat{Y}$ provides no additional information in predicting Y given either $\phi(\mathbf{X})$ or
 661 $\mathbf{X}$. This is because, the path $\hat{Y} \leftarrow \phi(\mathbf{X}) \leftarrow \mathbf{X} \leftrightarrow Y$, between $\hat{Y}$ and Y is blocked by either $\{\phi(\mathbf{X})\}$
 662 or $\{\mathbf{X}\}$. This implies $\hat{Y} \perp\!\!\!\perp Y|\phi(\mathbf{X})$ and $\hat{Y} \perp\!\!\!\perp Y|\mathbf{X}$. Now consider the expansion of $I(Y; \hat{Y})$.

$$\begin{aligned} I(Y; \hat{Y}) &= I(Y; \phi(\mathbf{X}), \hat{Y}) - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Introduce } \phi(\mathbf{X})) \\ &= I(\phi(\mathbf{X}); Y) + \underbrace{I(Y; \hat{Y}|\phi(\mathbf{X}))}_0 - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Expand and use } \hat{Y} \perp\!\!\!\perp Y | \phi(\mathbf{X})) \\ &= I(\phi(\mathbf{X}), E; Y) - I(Y; E|\phi(\mathbf{X})) - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Introduce } \mathbf{E}) \\ &= I(Y; E) + I(\phi(\mathbf{X}); Y|E) - I(Y; E|\phi(\mathbf{X})) - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Expand}) \end{aligned} \quad (6)$$

663 Similarly,

$$\begin{aligned} I(Y; \hat{Y}) &= I(Y; \phi(\mathbf{X}), \hat{Y}) - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Introduce } \phi(\mathbf{X})) \\ &= I(\phi(\mathbf{X}); Y) + \underbrace{I(Y; \hat{Y}|\phi(\mathbf{X}))}_0 - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Expand and use } \hat{Y} \perp\!\!\!\perp Y | \phi(\mathbf{X})) \\ &= I(Y, E; \phi(\mathbf{X})) - I(\phi(\mathbf{X}); E|Y) - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Introduce } \mathbf{E}) \\ &= I(\phi(\mathbf{X}); E) + I(\phi(\mathbf{X}); Y|E) - I(\phi(\mathbf{X}); E|Y) - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Expand}) \end{aligned} \quad (7)$$

664 Summing Equations (6) and (7) and grouping the terms gives the desired expression:

$$\begin{aligned} I(Y; \hat{Y}) &= I(\phi(\mathbf{X}); Y|E) - \frac{I(\phi(\mathbf{X}); E|Y)}{2} + \frac{I(Y; E)}{2} \\ &\quad + \frac{I(\phi(\mathbf{X}); E)}{2} - \frac{I(Y; E|\phi(\mathbf{X}))}{2} - I(\phi(\mathbf{X}); Y|\hat{Y}). \end{aligned}$$

665 To get the similar decomposition with $\mathbf{X}$ instead of $\phi(\mathbf{X})$, use $\hat{Y} \perp\!\!\!\perp Y | \mathbf{X}$ and introduce $\mathbf{X}$ into the
 666 expansion instead of $\phi(\mathbf{X})$. $\square$

667 **Proposition 4.2.** For a covariate vector $\mathbf{X}$, label Y , an environment variable E , a feature extractor
 668 ϕ , the prediction $\hat{Y}$, and an unobserved confounding variable U that cause both $\mathbf{X}$ and Y , and if
 669 either (i) $\mathbf{X} \rightarrow Y$ or (ii) $Y \rightarrow \mathbf{X}$, the predictive information $I(Y; \hat{Y})$ can be decomposed as follows:

$$I(Y; \hat{Y}) = \overbrace{I(\phi(\mathbf{X}); Y|E)}^{\text{Cond. informativeness}} - \overbrace{I(\phi(\mathbf{X}); Y|\hat{Y})}^{\text{Residual}} \quad (4)$$

670 *Proof.* Following the causal graph shown in Figure 3 of the main paper, we prove the proposition in
 671 two separate cases: $\mathbf{X} \rightarrow Y$ and $Y \rightarrow \mathbf{X}$.

672 **Case 1: $\mathbf{X} \rightarrow Y$.** In this case, we have the following inequality because conditioning on Y opens the
 673 path $\phi(\mathbf{X}) \leftarrow \mathbf{X} \rightarrow Y \leftarrow U \leftarrow E$ between $\phi(\mathbf{X})$ and E at the collider node Y :

$$I(\phi(\mathbf{X}); E|Y) \geq I(\phi(\mathbf{X}); E). \quad (8)$$

674 Additionally, considering the causal substructure: $Y \leftarrow \mathbf{X} \leftarrow U \leftrightarrow E$, we have the following
 675 inequality because conditioning on $\phi(\mathbf{X})$ partially blocks the information flow from Y to E at the
 676 node $\mathbf{X}$ as long as $\phi(\mathbf{X})$ encodes some information about $\mathbf{X}$:

$$I(Y; E) \geq I(Y; E|\phi(\mathbf{X})). \quad (9)$$

677 Now consider the following expansion from Equation (7) in the proof of Proposition 4.1.

$$\begin{aligned} I(Y; \hat{Y}) &= I(\phi(\mathbf{X}); E) + I(\phi(\mathbf{X}); Y|E) - I(\phi(\mathbf{X}); E|Y) - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Equation (7)}) \\ &\leq I(\phi(\mathbf{X}); Y|E) + I(\phi(\mathbf{X}); E|Y) - I(\phi(\mathbf{X}); E|Y) - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Using (8)}) \\ &= I(\phi(\mathbf{X}); Y|E) - I(\phi(\mathbf{X}); Y|\hat{Y}) \end{aligned}$$

678 Similarly, consider the following expansion from Equation (6) from the proof of Proposition 4.1:

$$\begin{aligned} I(Y; \hat{Y}) &= I(Y; E) + I(\phi(\mathbf{X}); Y|E) - I(Y; E|\phi(\mathbf{X})) - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Equation (6)}) \\ &\geq I(Y; E) + I(\phi(\mathbf{X}); Y|E) - I(Y; E) - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Using (9)}) \\ &= I(\phi(\mathbf{X}); Y|E) - I(\phi(\mathbf{X}); Y|\hat{Y}) \end{aligned}$$

679 That is, we have $I(\phi(\mathbf{X}); Y|E) - I(\phi(\mathbf{X}); Y|\hat{Y}) \leq I(Y; \hat{Y}) \leq I(\phi(\mathbf{X}); Y|E) - I(\phi(\mathbf{X}); Y|\hat{Y})$.

680 **Case 2: $Y \rightarrow \mathbf{X}$.** In this scenario, we have the following inequality because conditioning on $\phi(\mathbf{X})$
 681 opens the path $E \rightarrow U \rightarrow \mathbf{X} \leftarrow Y$ between Y and E at the collider node $\mathbf{X}$ because $\phi(\mathbf{X})$ is a child
 682 of $\mathbf{X}$ [58].

$$I(Y; E|\phi(\mathbf{X})) \geq I(Y; E) \quad (10)$$

683 Additionally, we have the following inequality because conditioning on Y blocks the path $\phi(\mathbf{X}) \leftarrow$
 684 $\mathbf{X} \leftarrow Y \leftarrow U \leftrightarrow E$ at Y and hence less information flow between $\phi(\mathbf{X})$ and E .

$$I(\phi(\mathbf{X}); E) \geq I(\phi(\mathbf{X}); E|Y) \quad (11)$$

685 Now consider the following expansion from Equation (6) from the proof of Proposition 4.1.

$$\begin{aligned} I(Y; \hat{Y}) &= I(Y; E) + I(\phi(\mathbf{X}); Y|E) - I(Y; E|\phi(\mathbf{X})) - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Equation (6)}) \\ &\leq I(\phi(\mathbf{X}); Y|E) + I(Y; E|\phi(\mathbf{X})) - I(Y; E|\phi(\mathbf{X})) - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Using (10)}) \\ &= I(\phi(\mathbf{X}); Y|E) - I(\phi(\mathbf{X}); Y|\hat{Y}) \end{aligned}$$

686 Similarly, consider the following expansion from Equation (7) from the proof of Proposition 4.1.

$$\begin{aligned} I(Y; \hat{Y}) &= I(\phi(\mathbf{X}); E) + I(\phi(\mathbf{X}); Y|E) - I(\phi(\mathbf{X}); E|Y) - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Equation (7)}) \\ &\geq I(\phi(\mathbf{X}); Y|E) + I(\phi(\mathbf{X}); E) - I(\phi(\mathbf{X}); E) - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Using (11)}) \\ &= I(\phi(\mathbf{X}); Y|E) - I(\phi(\mathbf{X}); Y|\hat{Y}) \end{aligned}$$

687 That is, we have $I(\phi(\mathbf{X}); Y|E) - I(\phi(\mathbf{X}); Y|\hat{Y}) \leq I(Y; \hat{Y}) \leq I(\phi(\mathbf{X}); Y|E) - I(\phi(\mathbf{X}); Y|\hat{Y})$.

688 Since the upper bound and lower bounds are the same in both cases, we have the desired result
 689 $I(Y; \hat{Y}) = I(\phi(\mathbf{X}); Y|E) - I(\phi(\mathbf{X}); Y|\hat{Y})$. $\square$

690 **Proposition 4.3.** If $\phi_1(\cdot), \phi_2(\cdot)$ are invertible functions and if $\mathbf{X}_I$ are informative variables such that
 691 $Y \not\perp\!\!\!\perp \mathbf{X}_I \mid \mathbf{X}, E$, then we have the following inequalities:

$$\begin{aligned} (i) & I(\phi_2(\{\mathbf{X} \cup \mathbf{X}_I\}); Y|E) > I(\phi_1(\mathbf{X}); Y|E) \quad (ii) I(\phi_2(\{\mathbf{X} \cup \mathbf{X}_I\}); E) > I(\phi_1(\mathbf{X}); E) \\ (iii) & I(Y; E|\phi_2(\{\mathbf{X} \cup \mathbf{X}_I\})) < I(Y; E|\phi_1(\mathbf{X})) \quad (iv) I(\phi_2(\{\mathbf{X} \cup \mathbf{X}_I\}); E \mid Y) > I(\phi_1(\mathbf{X}); E \mid Y) \end{aligned} \quad (5)$$

692 *Proof.* (i) By the chain rule of mutual information, we have: $I(\mathbf{X}, \mathbf{X}_I; Y|E) = I(\mathbf{X}; Y|E) +$
693 $I(\mathbf{X}_I; Y|E, \mathbf{X})$. We have $I(\mathbf{X}_I; Y|E, \mathbf{X}) > 0$ because $Y \not\perp\!\!\!\perp \mathbf{X}_I \mid \mathbf{X}, E$. Hence, we have
694 $I(\mathbf{X}, \mathbf{X}_I; Y|E) > I(\mathbf{X}; Y|E)$. Since $\phi_1(\cdot), \phi_2(\cdot)$ are invertible functions and mutual information is
695 invariant to invertible transformations, we have $I(\phi_2(\{\mathbf{X} \cup \mathbf{X}_I\}); Y|E) > I(\phi_1(\mathbf{X}); Y|E)$.

696 (ii) We consider the causal graph shown in Figure 3. From the chain rule of mutual information,
697 we have: $I(\mathbf{X}, \mathbf{X}_I; E) = I(\mathbf{X}; E) + I(\mathbf{X}_I; E|\mathbf{X})$. Since $E \leftrightarrow U \rightarrow Y$, any covariate that is
698 informative to Y is also informative to U . Since $E \leftrightarrow U$, we have $I(\mathbf{X}_I; E|\mathbf{X}) > 0$, and hence
699 $I(\mathbf{X}, \mathbf{X}_I; E) > I(\mathbf{X}; E)$. Since $\phi_1(\cdot), \phi_2(\cdot)$ are invertible functions and mutual information is
700 invariant to invertible transformations, we have $I(\phi_2(\{\mathbf{X} \cup \mathbf{X}_I\}); E) > I(\phi_1(\mathbf{X}); E)$.

701 (iii) Consider $I(Y; E, \mathbf{X}_I | \mathbf{X})$. It can be expressed in two ways:

$$I(Y; E, \mathbf{X}_I | \mathbf{X}) = I(Y; E | \mathbf{X}) + I(Y; \mathbf{X}_I | E, \mathbf{X}) = I(Y; \mathbf{X}_I | \mathbf{X}) + I(Y; E | \mathbf{X}_I, \mathbf{X})$$

702 Equating and rearranging the terms gives the following.

$$I(Y; E | \mathbf{X}) - I(Y; E | \mathbf{X}_I, \mathbf{X}) = I(Y; \mathbf{X}_I | \mathbf{X}) - I(Y; \mathbf{X}_I | E, \mathbf{X})$$

703 Since $Y \not\perp\!\!\!\perp \mathbf{X}_I \mid E, \mathbf{X}$, we have $I(Y; \mathbf{X}_I | \mathbf{X}) > I(Y; \mathbf{X}_I | E, \mathbf{X})$ because conditioning usually
704 reduces mutual information unless the additional conditioning variable opens any collider paths. Here
705 E is neither a collider nor a descendant of any collider. Thus, $I(Y; E | \mathbf{X}) > I(Y; E | \mathbf{X}_I, \mathbf{X})$. Since
706 $\phi_1(\cdot), \phi_2(\cdot)$ are invertible functions and mutual information is invariant to invertible transformations,
707 we have $I(Y; E | \phi_1(\mathbf{X})) > I(Y; E | \phi_2(\{\mathbf{X}_I \cup \mathbf{X}\}))$.

708 (iv) By the chain rule of mutual information, we have: $I(\mathbf{X}, \mathbf{X}_I; E | Y) = I(\mathbf{X}; E | Y) +$
709 $I(\mathbf{X}_I; E|\mathbf{X}, Y)$. Since $E \leftrightarrow U \rightarrow Y$, any covariate that is informative to Y is also informative to
710 U . Since $E \leftrightarrow U$, we have $I(\mathbf{X}_I; E|\mathbf{X}, Y) > 0$, and hence $I(\mathbf{X}, \mathbf{X}_I; E | Y) > I(\mathbf{X}; E | Y)$. Since
711 $\phi_1(\cdot), \phi_2(\cdot)$ are invertible functions and mutual information is invariant to invertible transformations,
712 we have $I(\phi_2(\{\mathbf{X} \cup \mathbf{X}_I\}); E | Y) > I(\phi_1(\mathbf{X}); E | Y)$. $\square$

713 B Experimental setup

714 This section details our experimental setup. Random seeds have a statistically insignificant impact
715 on the results in most settings [24]. For the main-paper results, we report the mean and standard
716 deviation across five random hyperparameter values (sampled from the domains in Table B1) for each
717 dataset–method combination. To evaluate the mutual information terms, we use the nonparametric
718 entropy estimation toolbox [37, 77, 78], and, following [24], we compute mutual information on both
719 in-distribution (ID) test data (from training domains) and out-of-distribution (OOD) test data (from
720 test domains).

721 For consistency, we draw 20,000 random samples
722 from each of the ID and OOD test sets,
723 though some datasets contain fewer than 20,000
724 samples. Sample size has a negligible effect
725 on the evaluated mutual information terms: as
726 Figure B1 shows, the terms in Equation (1) com-
727 puted for the XGB representations on the *in-*
728 *come* dataset remain stable as sample size varies.
729 Features $\phi(\mathbf{X})$ are extracted from the layer pre-
730 ceeding the classification head; for XGBoost, we
731 combine the model’s output margins and SHAP
732 values. All experiments were run on a single
733 NVIDIA RTX A6000 GPU.

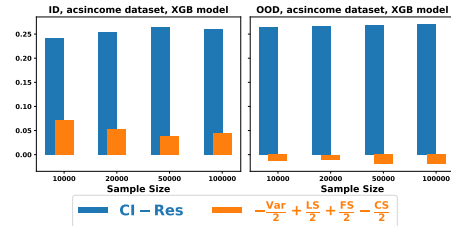


Figure B1: The evaluated terms of predictive information stay consistent across sample sizes on the *income* dataset.

734 For the experiments on the impact of informative covariates, we follow the categorization of Nastl
735 and Hardt [57], partitioning covariates into *causal*, *arguably causal*, and *all covariates*. Causal
736 covariates have a clearly established influence on the target, with well-supported directionality
737 and unlikely reverse causation—though restricting to them risks omitting relevant parents due
738 to knowledge gaps. Arguably causal covariates have uncertain causal relationships, satisfying
739 at least one of: (1) being known causal, (2) having plausible but potentially bidirectional influ-
740 ence, or (3) a likely but unconfirmed causal effect. These groups approximate the true causal
741 parents given available knowledge, though relationships may be confounded. Some datasets

Table B1: Hyperparameters and their possible values considered in this paper.

Model	Hyperparameter	Search values
XGBoost	Learning rate	[0.1, 0.2, 0.3]
	Maximum tree depth	[4, 5, 6]
	Minimum child weight	[0.1, 1, 10]
	Gamma (min. loss reduction)	[0.0001, 0.001, 0.01]
	Subsample ratio	[0.5, 0.6, 0.7]
	Column subsample per tree	[0.5, 0.6, 0.7]
	L1 regularization (α)	[0.0001, 0.001, 0.01]
	L2 regularization (λ)	[0.0001, 0.001, 0.01]
MLP	Number of layers	[2, 3, 4]
	Hidden layer size	[256, 512, 1024]
	Dropout rate	[0.0, 0.1, 0.2]
	Learning rate	[0.01, 0.02, 0.05]
	Weight decay (L2)	[0.0001, 0.001, 0.01]
	Batch size	[4096]
	Number of epochs	[1, 2, 3]
GroupDRO	Number of layers	[2, 3, 4]
	Hidden layer size	[256, 512, 1024]
	Group-weights step size	[0.01, 0.05, 0.1]
	Dropout rate	[0.0, 0.1, 0.2]
	Learning rate	[0.01, 0.02, 0.05]
	Weight decay (L2)	[0.0001, 0.001, 0.01]
	Batch size	[4096]
IRM	Number of layers	[2, 3, 4]
	Hidden layer size	[256, 512, 1024]
	Dropout rate	[0.0, 0.1, 0.2]
	IRM penalty weight (λ)	[0.01, 0.05, 0.1]
	IRM penalty anneal iterations	[1, 2, 3]
	Learning rate	[0.01, 0.02, 0.05]
	Weight decay (L2)	[0.0001, 0.001, 0.01]
	Batch size	[4096]
VREX	Number of layers	[2, 3, 4]
	Hidden layer size	[256, 512, 1024]
	VREX penalty anneal iterations	[1, 2, 3]
	VREX penalty weight (λ)	[0.1, 10, 100]
	Dropout rate	[0.0, 0.1, 0.2]
	Learning rate	[0.01, 0.02, 0.05]
	Weight decay (L2)	[0.0001, 0.001, 0.01]
	Batch size	[4096]
VREX	Number of layers	[2, 3, 4]
	Hidden layer size	[256, 512, 1024]
	VREX penalty anneal iterations	[1, 2, 3]
	VREX penalty weight (λ)	[0.1, 10, 100]
	Dropout rate	[0.0, 0.1, 0.2]
	Learning rate	[0.01, 0.02, 0.05]
	Weight decay (L2)	[0.0001, 0.001, 0.01]
	Batch size	[4096]
VREX	Number of layers	[2, 3, 4]
	Hidden layer size	[256, 512, 1024]
	VREX penalty anneal iterations	[1, 2, 3]
	VREX penalty weight (λ)	[0.1, 10, 100]
	Dropout rate	[0.0, 0.1, 0.2]
	Learning rate	[0.01, 0.02, 0.05]
	Weight decay (L2)	[0.0001, 0.001, 0.01]
	Batch size	[4096]

also contain *anti-causal* covariates, which the target likely affects but not vice versa. The *all covariates* set contains every observed variable, regardless of causal status. The sign consistency metric from the main paper § 5 is defined as follows. For each $(m, \sigma_m) \in M$ where $M = \{(\text{conditional informativeness}, +1), (\text{variation}, -1), (\text{label shift}, +1), (\text{feature shift}, +1), (\text{concept shift}, -1), \text{ and } (\text{residual}, -1)\}$, for a specific dataset, we compute:

$$C_m = \frac{1}{2|P|} \sum_{(s,t) \in P} [\mathbb{I}(\sigma_m \cdot (\mu_t^{\text{tr}}(m) - \mu_s^{\text{tr}}(m)) > 0) + \mathbb{I}(\sigma_m \cdot (\mu_t^{\text{te}}(m) - \mu_s^{\text{te}}(m)) > 0)]$$

Where P is the set of covariate set pairs (s, t) with $t \supset s$. For example, s can be causal covariates and t can be arguably causal covariates. There is a factor of 2 in $2P$ that accounts for both train and

test data. $\mu_s^{tr/te}(m)$ is the mean of measure m for covariate set s in training/testing data. $\mathbb{I}(\cdot)$ is the indicator function. $|P| = 3$ for covariate settings $\{causal, arguably\ causal, all\}$. To get final sign consistency metric value for each m , we average across datasets: $\bar{C}_m = \frac{1}{|D|} \sum_{d \in D} C_m^{(d)}$ where $|D|$ is the number of datasets.

C Additional results on real-world datasets

This section presents additional results on real-world datasets. Figures C2 and C3 compare the terms *conditional informativeness* – *residual* and –*variation* / 2 + *label shift* / 2 + *feature shift* / 2 – *concept shift* / 2 for each dataset (aggregated in the main paper’s Figure 4 Left). Two conclusions emerge. By Proposition 4.1, the sum of the blue and orange terms is positively correlated with overall model performance. Moreover, because of hidden confounding shift in real-world data, the *conditional informativeness* – *residual* difference contributes substantially more to the overall predictive information than –*variation* / 2 + *label shift* / 2 + *feature shift* / 2 – *concept shift* / 2 does. This is direct motivation to maximize conditional informativeness for better OOD generalization under hidden confounding shift.

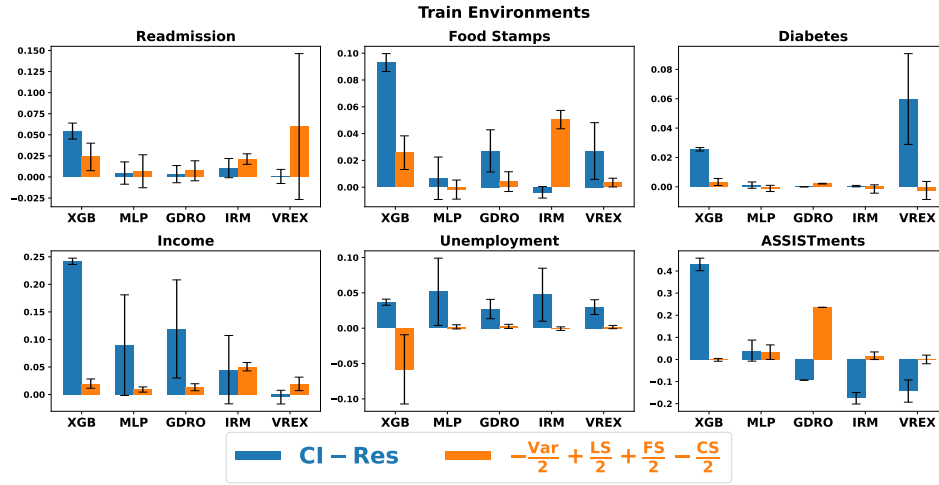


Figure C2: Dataset-specific results on ID data.

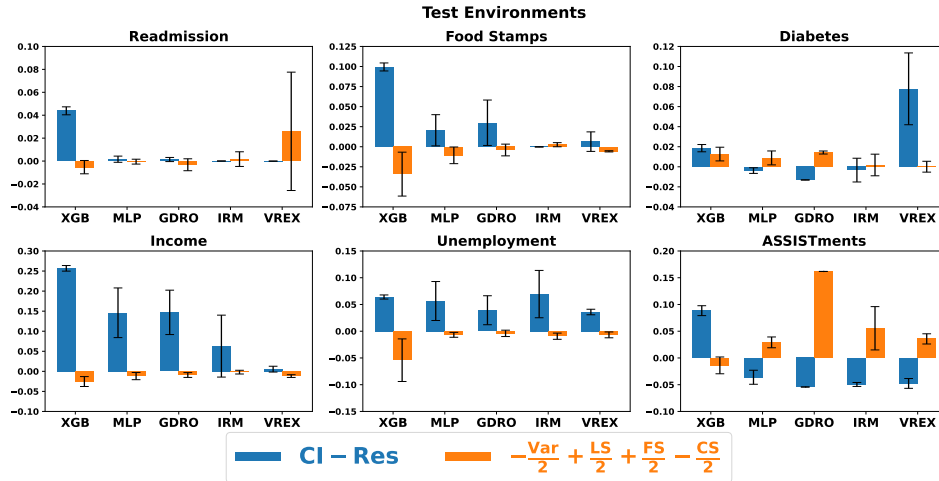


Figure C3: Dataset-specific results on OOD data.

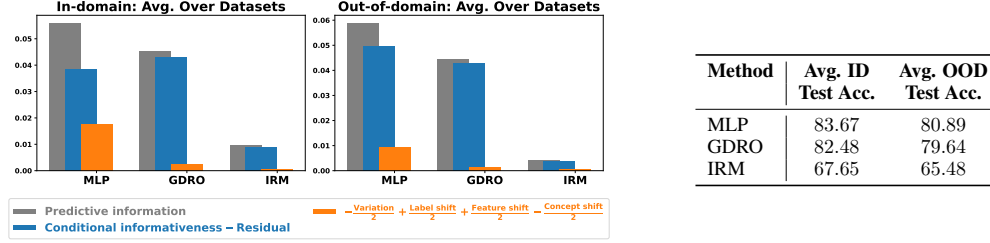


Figure C4: The difference *conditional informativeness* – *residual* is positively correlated with the average ID and OOD test accuracy over the five datasets in the table (right). This difference contributes substantially more to the overall predictive information than $-\text{variation} / 2 + \text{label shift} / 2 + \text{feature shift} / 2 - \text{concept shift} / 2$ does.

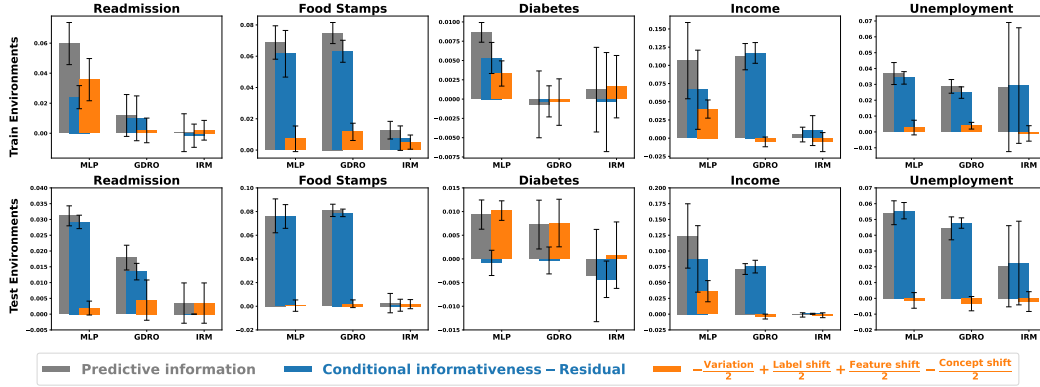


Figure C5: Dataset-specific comparison of the terms in the predictive information.

Table C2: ID and OOD test accuracies of the models across datasets.

Method	Readmission		Food stamps		Diabetes	
	ID test acc.	OOD test acc.	ID test acc.	OOD test acc.	ID test acc.	OOD test acc.
MLP	65.78 ± 0.14	61.75 ± 0.22	84.74 ± 0.03	82.00 ± 0.06	87.66 ± 0.02	83.22 ± 0.05
GDRO	60.47 ± 1.08	57.60 ± 0.42	84.47 ± 0.04	81.41 ± 0.09	87.47 ± 0.10	82.90 ± 0.10
IRM	50.28 ± 7.35	50.85 ± 1.74	80.91 ± 0.00	78.01 ± 0.00	42.54 ± 36.5	43.48 ± 31.9
	Income		Unemployment			
	ID test acc.	OOD test acc.	ID test acc.	OOD test acc.		
MLP	82.92 ± 0.05	81.46 ± 0.01	97.27 ± 0.02	96.04 ± 0.05		
GDRO	82.70 ± 0.05	80.16 ± 0.45	97.28 ± 0.01	96.10 ± 0.00		
IRM	67.91 ± 0.00	60.20 ± 0.00	96.61 ± 0.00	94.84 ± 0.00		

Figures C6–C15 compare the individual terms of the information decomposition in (1) for each method and dataset. The pattern is clear and consistent. The second column (*variation*) and third column (*feature shift*) are similar in magnitude, so by Equation 1 their difference is near zero and contributes little to the decomposition. The fourth column (*label shift*) and fifth column (*concept shift*) are likewise similar, and their difference is again near zero. As a result, in almost every setting the difference *conditional informativeness* – *residual* is the dominant contributor to the predictive information.

Hyperparameter tuning: We also perform hyperparameter tuning, running a grid search and selecting the best values by ID validation accuracy (not OOD validation accuracy). This amounts to 1500 experiments (5 datasets × 3 models × 100 hyperparameter choices sampled from a grid). To obtain means and standard deviations for accuracies and mutual information terms, we repeat each experiment over 5 random seeds, even though their effect is insignificant in most cases. The results (Figure C4) reinforce the main-paper findings: by Proposition 4.1, the sum of the *blue* and *orange*

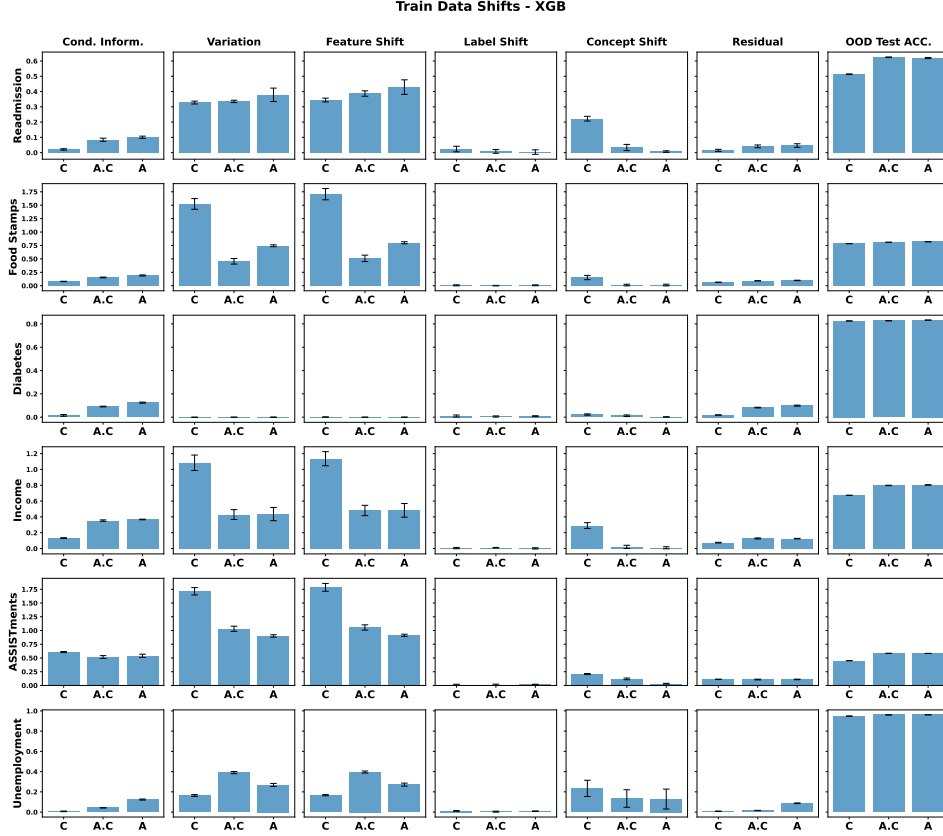


Figure C6: Decomposition of information metrics on train data for XGBoost and groups of features: (C) Causal, (A.C) Arguably causal, (A) All.

terms is positively correlated with overall model performance, and, owing to hidden confounding shift in real-world data, the *conditional informativeness* – *residual* difference contributes far more to the predictive information than $-\text{variation} / 2 + \text{label shift} / 2 + \text{feature shift} / 2 - \text{concept shift} / 2$. The dataset-specific results in Figure C5 and Table C2 follow the same trend.

Scope of the predictive information decomposition analysis: Predictive information $I(Y; \hat{Y})$ measures the statistical dependence between predictions and labels, not accuracy itself. A binary classifier that systematically predicts the *wrong* label (e.g., $\hat{Y} = 1$ when $Y = 0$) attains the maximal $I(Y; \hat{Y}) = 1$ bit through perfect anti-correlation yet achieves 0% accuracy, because mutual information quantifies *reduction in uncertainty* rather than correctness. The other terms sharpen this picture: conditional informativeness $I(\phi(X); Y|E)$ can mask harmful variation $I(\phi(X); E|Y)$, and feature shift $I(\phi(X); E)$ can inflate $I(Y; \hat{Y})$ through spurious correlations. These caveats do not undercut the analysis—for the reasonably performant models our experiments target, the decomposition reliably reveals whether predictive information comes from generalizable patterns or environment-specific artifacts, enabling targeted, interpretable improvements.

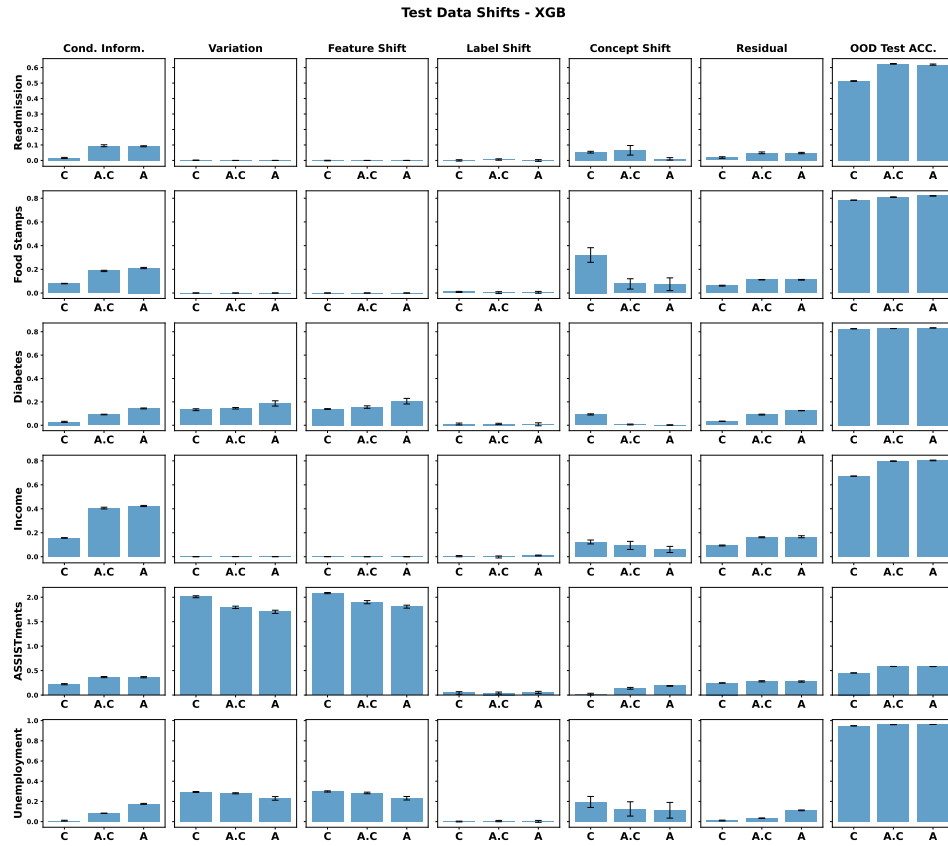


Figure C7: Decomposition of information metrics on test data for XGBoost and groups of features: (C) Causal, (A.C) Arguably causal, (A) All.

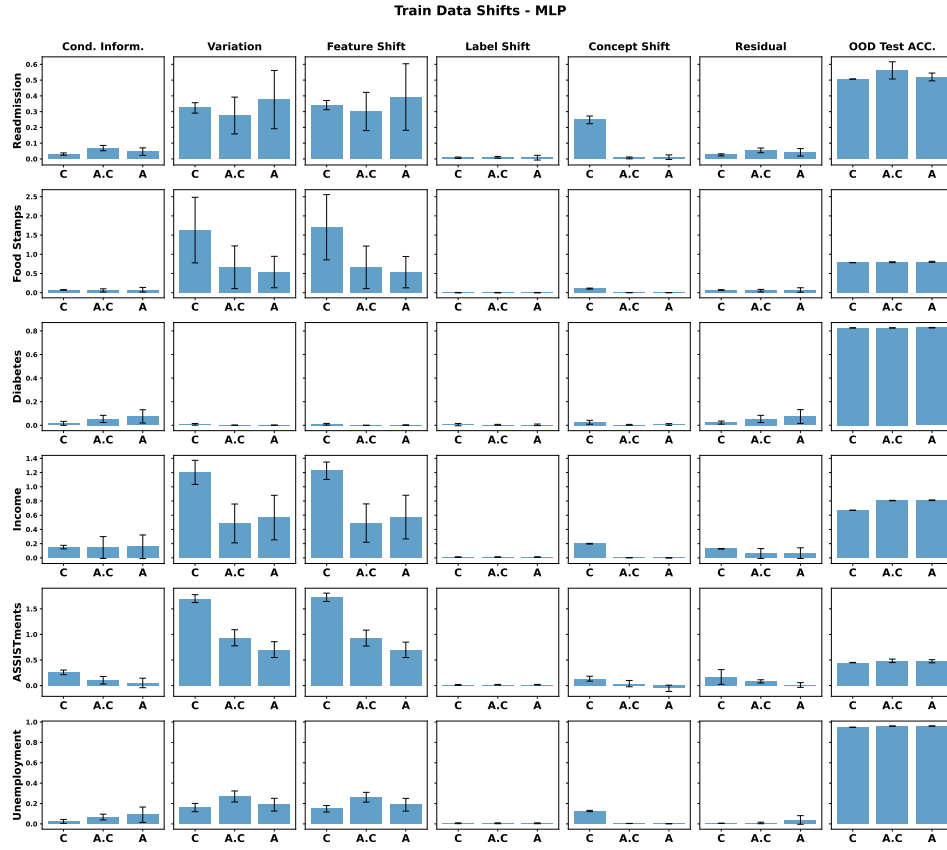


Figure C8: Decomposition of information metrics on train data for MLP and groups of features: (C) Causal, (A.C) Arguably causal, (A) All.

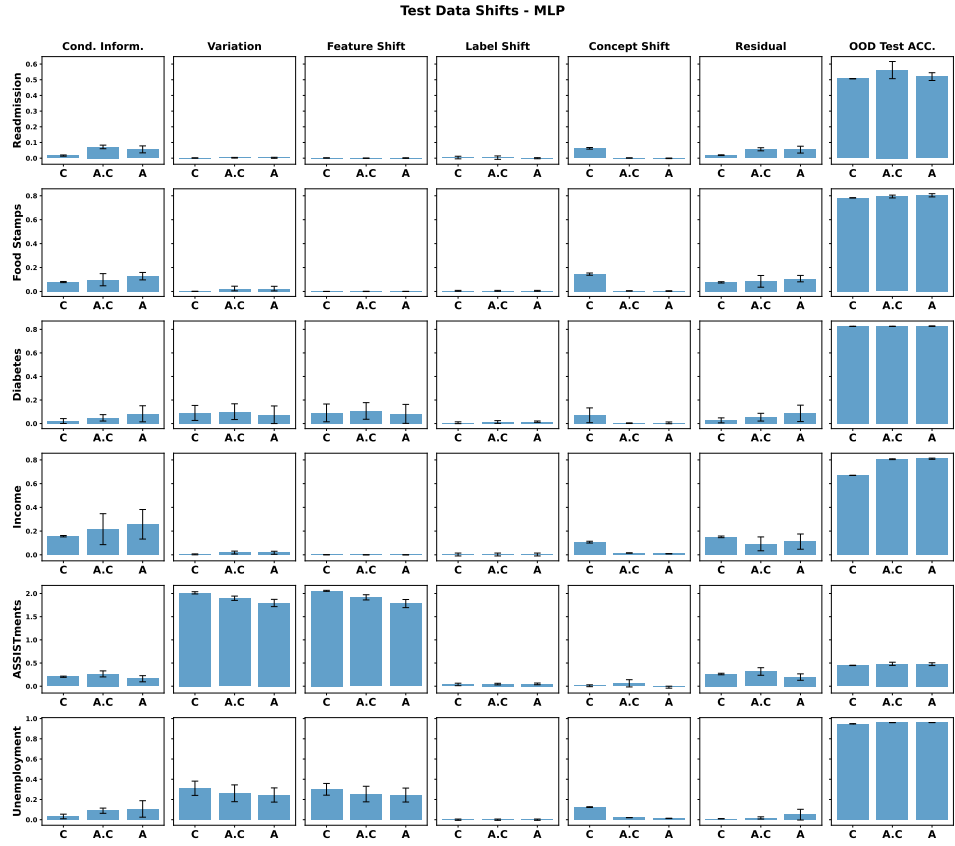


Figure C9: Decomposition of information metrics on test data for MLP and groups of features: (C) Causal, (A.C) Arguably causal, (A) All.

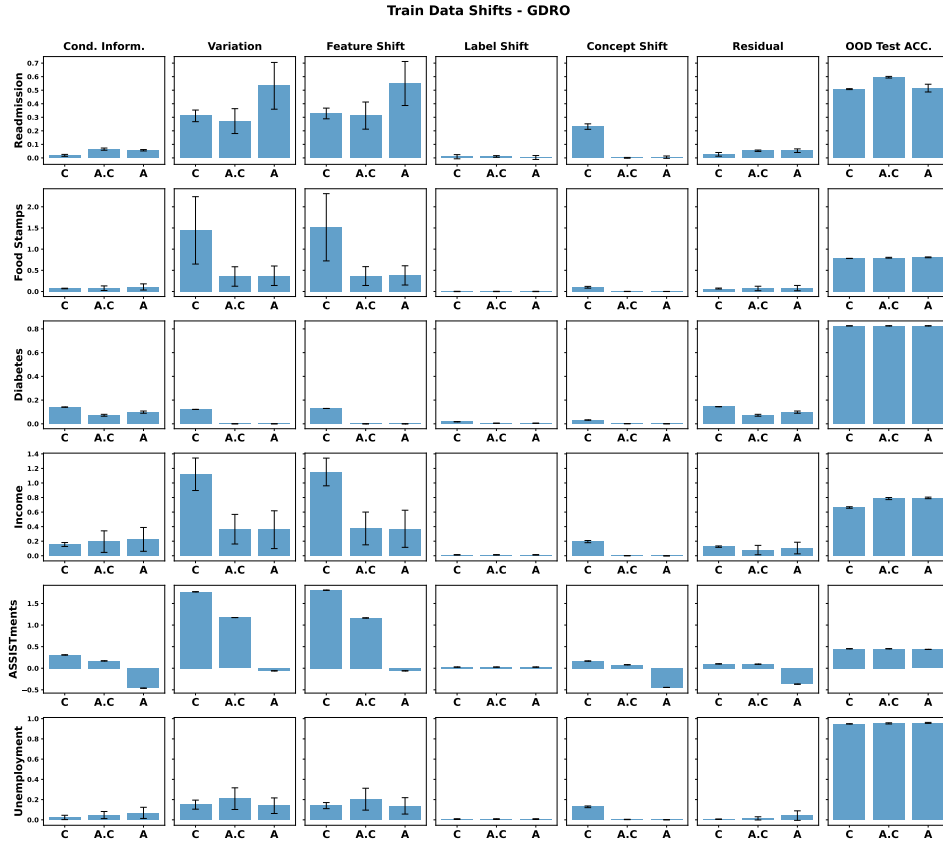


Figure C10: Decomposition of information metrics on train data for GDRO and groups of features: (C) Causal, (A.C) Arguably causal, (A) All.

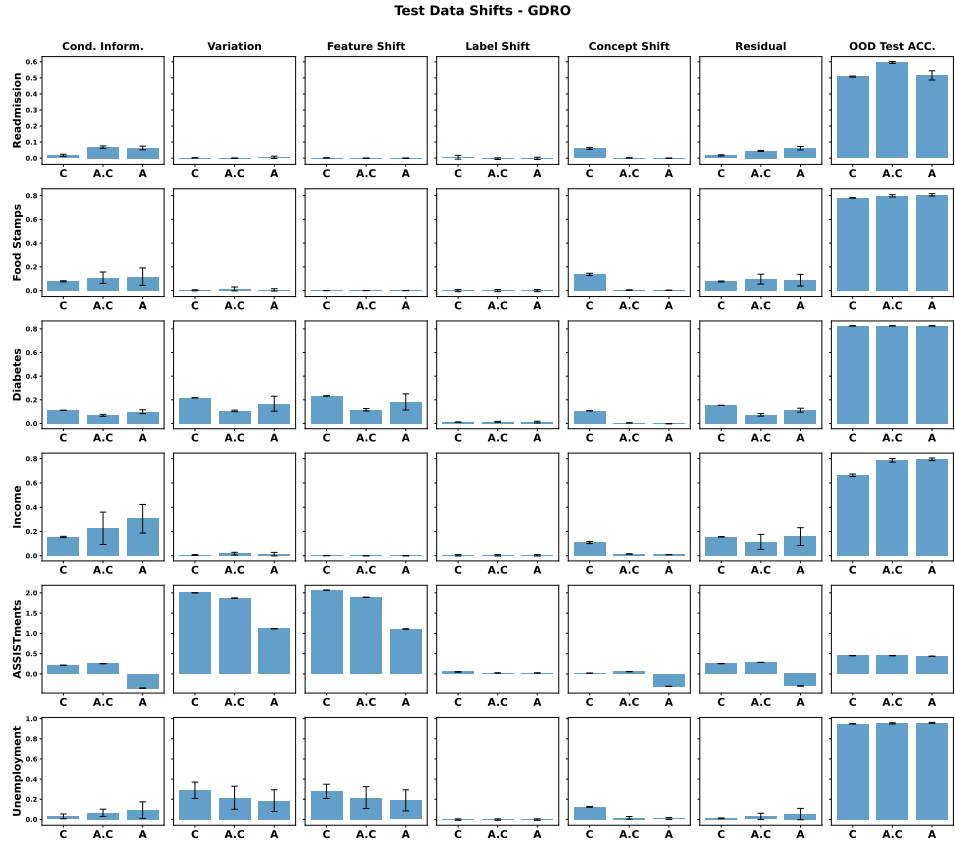


Figure C11: Decomposition of information metrics on test data for GDRO and groups of features: (C) Causal, (A.C) Arguably causal, (A) All.

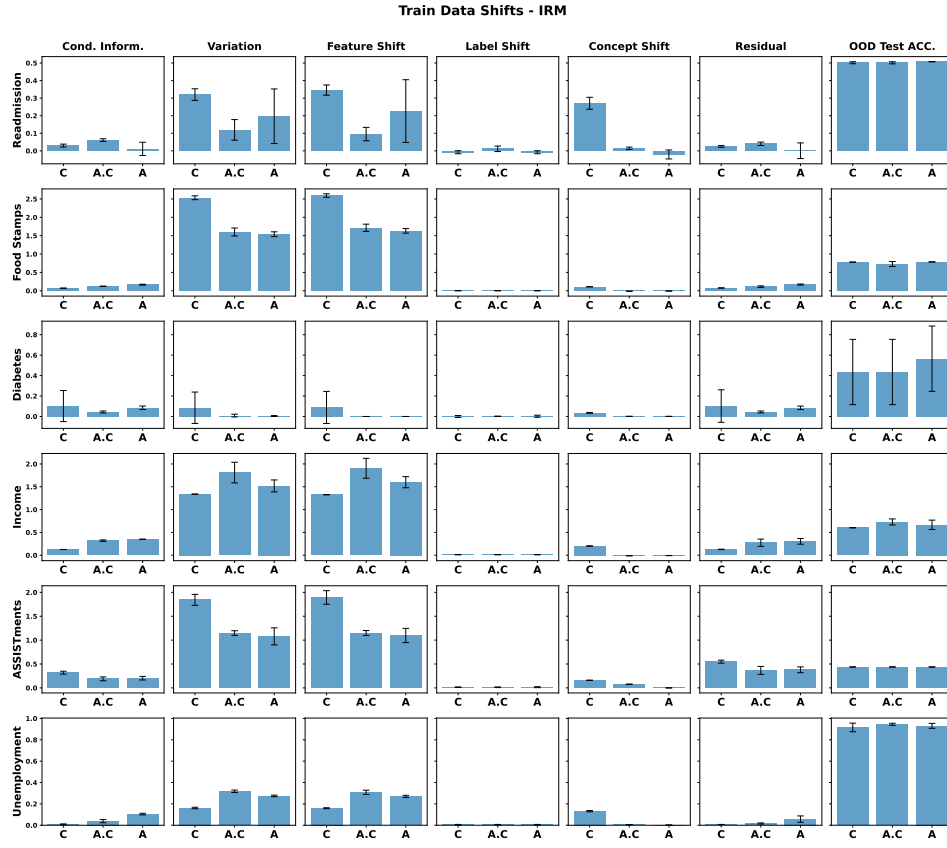


Figure C12: Decomposition of information metrics on train data for IRM and groups of features: (C) Causal, (A.C) Arguably causal, (A) All.

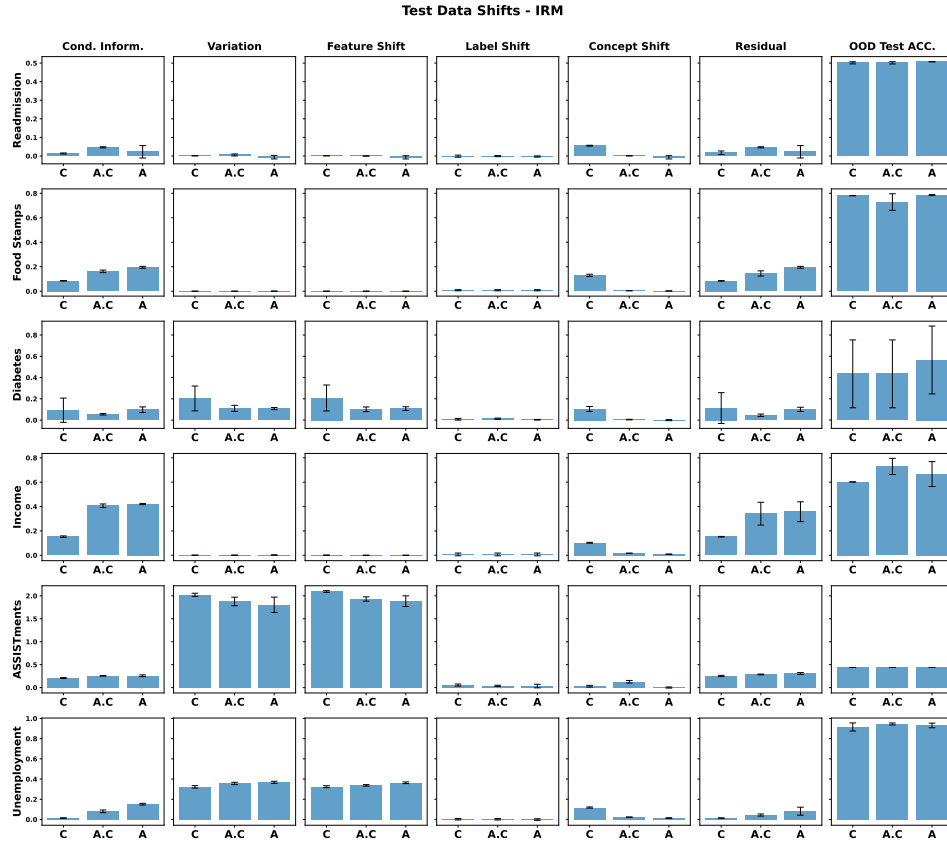


Figure C13: Decomposition of information metrics on test data for IRM and groups of features: (C) Causal, (A.C) Arguably causal, (A) All.

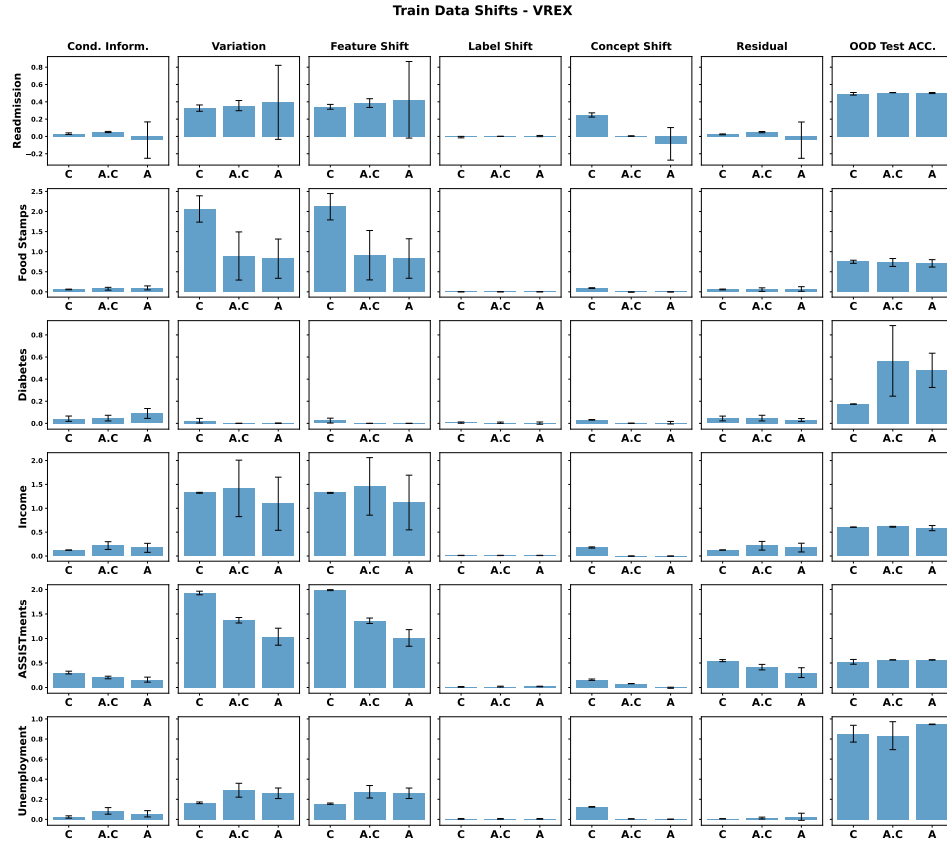


Figure C14: Decomposition of information metrics on train data for VREX and groups of features: (C) Causal, (A.C) Arguably causal, (A) All.

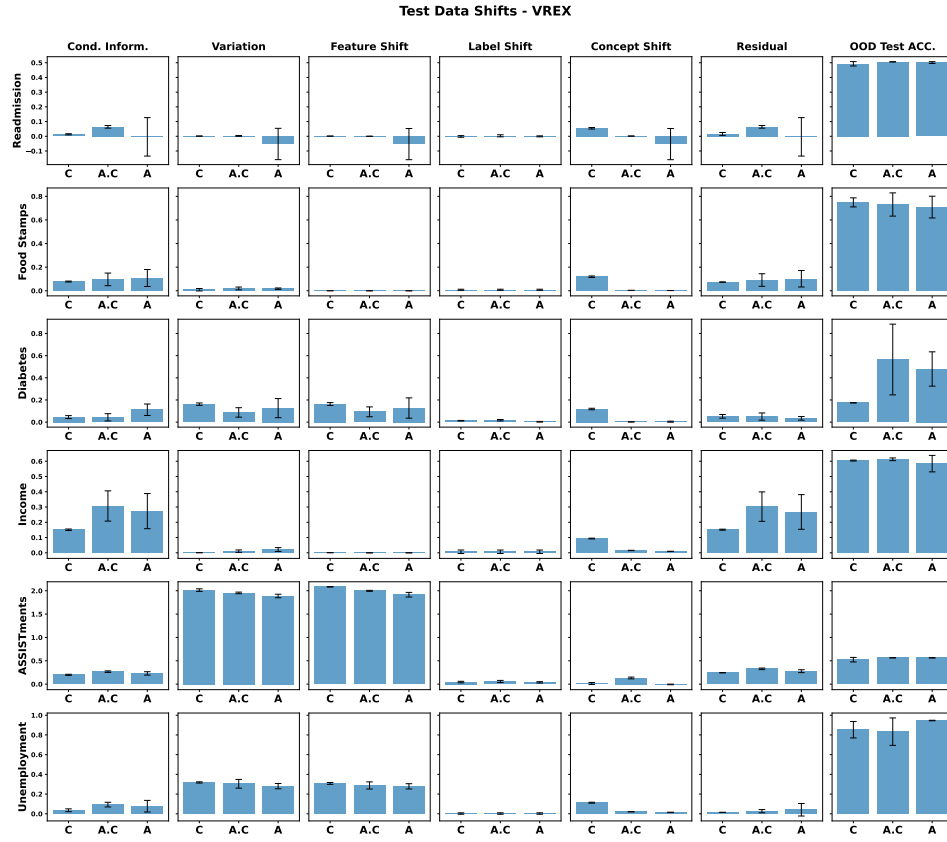


Figure C15: Decomposition of information metrics on test data for VREX and groups of features: (C) Causal, (A.C) Arguably causal, (A) All.

790 D Results on synthetic datasets

791 We run several synthetic experiments to establish: (i) the impact of different informativeness criteria,
 792 (ii) how the level of confounder overlap across environments affects performance, and (iii) the
 793 influence of the number of causal versus anti-causal variables. The data-generating process follows
 794 the structural equations in (12), with causal structure $U \rightarrow X, U \rightarrow Y, U \rightarrow \mathbf{X}_I$, and $X \rightarrow Y$,
 795 where U is an unobserved confounder, X an observed covariate, Y the target, and $\mathbf{X}_I$ the additional
 796 informative covariates.

$$\begin{aligned} U &\sim \mathcal{N}(\mu_u^e, \sigma_u) & X &\leftarrow f_X(U) + \mathcal{N}(\mu_x, \sigma_x) \\ X_i &\leftarrow f_i(U) + \mathcal{N}(\mu_i, \sigma_i); X_i \in \mathbf{X}_I & Y &\leftarrow f_Y(X, U) + \mathcal{N}(\mu_y, \sigma_y) \end{aligned} \quad (12)$$

797 We explore several functional forms for f_X, f_i , and f_Y . Domain shifts are induced by systematically
 798 varying the environment-specific mean μ_u^e of the confounder distribution, letting us control exactly
 799 how much distributional shift each environment sees and measure its effect on performance.

800 **Informativeness vs. accuracy:** We begin with linear structural equations $f_X = 0.3U$, $f_i =$
 801 $0.1U$, and $f_Y = X - 2U$, with $\mu_u^e \in \{-2, 2\}$ for training environments and $\mu_u^e \in \{0, 4\}$ for test
 802 environments. For Figure 5 of the main paper, we use one hidden confounder with $|\mathbf{X}_I| = 20$
 803 informative covariates. Increasing the number of informative covariates from 0 to 20 produces
 804 exactly the predicted movement: (i) lower mean squared error, (ii) higher conditional informativeness,
 805 (iii) higher feature shift, and (iv) lower concept shift. Two further informativeness criteria confirm
 806 the same behavior. First, with one hidden confounder and a single informative covariate ($|\mathbf{X}_I| = 1$)
 807 across noise levels $\sigma_i \in \{0, 0.1, \dots, 2.0\}$, Figure D16 shows that decreasing noise from 2.0 to 0.0
 808 again yields (i) lower MSE, (ii) higher conditional informativeness, (iii) higher feature shift, and
 809 (iv) lower concept shift, with the variation term fixed at zero as in all our synthetic experiments.
 810 Second, with 20 hidden confounders and 20 corresponding informative covariates under $f_X = 0.3U$,
 811 $f_i = 0.2U$, and $f_Y = X - 2U$, Figure D17 reproduces the identical pattern. Across all three criteria,
 812 adding informativeness reliably improves OOD generalization.

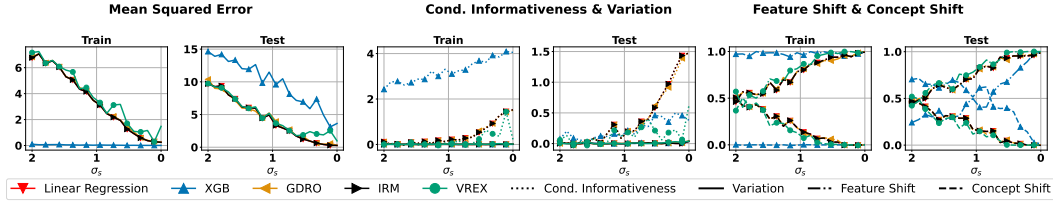


Figure D16: A low-noise proxy variable reduces MSE, raises conditional informativeness and feature shift, and lowers concept shift.

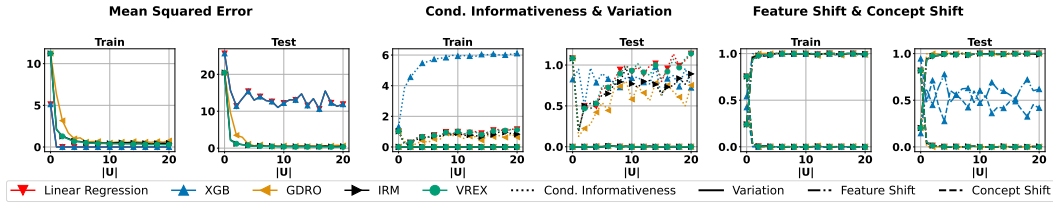


Figure D17: Adding proxy variables $\mathbf{X}_I$ of a set of hidden confounders $\mathbf{U}$ that are informative of Y reduces MSE, raises conditional informativeness and feature shift, and lowers concept shift.

813 **Overlapping confounder support vs. accuracy:** As noted in the main paper, the method of Prashant
 814 et al. [62] for OOD generalization under hidden confounding shift assumes the test confounder
 815 support stays within the training support. Having shown that correct relationships are learnable
 816 in synthetic linear data when the observed covariates carry enough information about the hidden
 817 confounders, we now extend the analysis to nonlinear data with MLP and XGB models under both
 818 low and high confounding support (Figures D18–D21). Under low confounding overlap (distant μ_e
 819 between ID and OOD data), models relying solely on X learn an inadequate global function that

Structural Equations :

(Hidden Confounder) $U \sim \mathcal{N}(\mu_e, 0.5^2)$
 $\mu_e \in \{-2, 2\}$ for ID
 $\mu_e \in \{0, 4\}$ for OOD
(Input) $X = 0.75U + \mathcal{N}(0, 1.5^2)$
(Outcome) $Y = X^2 - 3U + \mathcal{N}(0, 0.5^2)$
(Informative) $X_i = 0.6U + \mathcal{N}(0, 0.3^2)$

▲ Train data
▼ Test data
★ Predictions on train data
✱ Predictions on test data

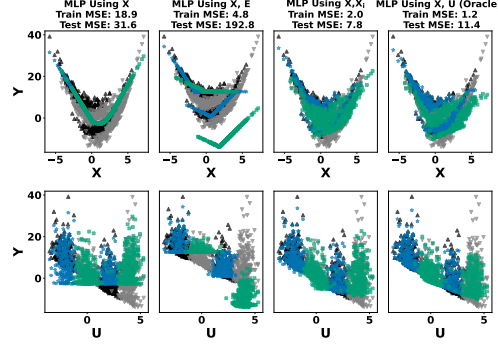


Figure D18: Performance of MLP on the low overlap non-linear setting, for sets of features: (i) X (ii) X and E (environment statistics) (iii) X and X_i (iv) X and U (Oracle).

Structural Equations :

(Hidden Confounder) $U \sim \mathcal{N}(\mu_e, 0.5^2)$
 $\mu_e \in \{-1, 0\}$ for ID
 $\mu_e \in \{-0.5, 0.5\}$ for OOD
(Input) $X = 0.75U + \mathcal{N}(0, 1.5^2)$
(Outcome) $Y = X^2 - 3U + \mathcal{N}(0, 0.5^2)$
(Informative) $X_i = 0.6U + \mathcal{N}(0, 0.3^2)$

▲ Train data
▼ Test data
★ Predictions on train data
✱ Predictions on test data

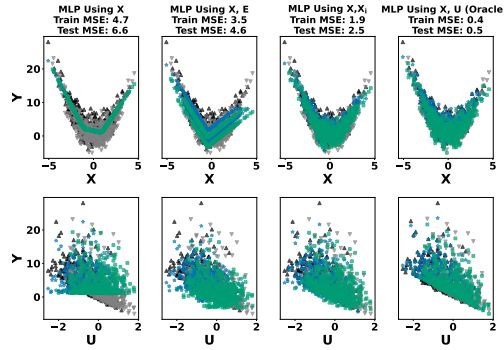


Figure D19: Performance of MLP on the high overlap non-linear setting, for sets of features: (i) X (ii) X and E (environment statistics) (iii) X and X_i (iv) X and U (Oracle).

cannot distinguish environments. Adding environment-specific summary statistics of the observed covariates captures environment-specific relationships and improves performance, but the largest gains come only from additional informative covariates—precisely as our theory predicts. For completeness, we include oracle results and U - Y scatter plots that expose the learned input-output mappings. As expected, performance is stronger under high confounding overlap; more importantly, the results in the low-overlap regime are strong too—a previously unstudied setting that directly challenges the common-support assumption in the literature.

Number of causal vs anti-causal variables: Recent benchmarks [24] show that baseline methods (e.g., XGB, MLP) consistently match or outperform OOD generalization techniques (e.g., GDRO, IRM, VREX) on real-world data, coinciding with the prevalent causal structure $X \rightarrow Y$ in tabular

Structural Equations :

(Hidden Confounder) $U \sim \mathcal{N}(\mu_e, 0.5^2)$
 $\mu_e \in \{-2, 2\}$ for ID
 $\mu_e \in \{0, 4\}$ for OOD
(Input) $X = 0.75U + \mathcal{N}(0, 1.5^2)$
(Outcome) $Y = X^2 - 3U + \mathcal{N}(0, 0.5^2)$
(Informative) $X_i = 0.6U + \mathcal{N}(0, 0.3^2)$

▲ Train data
▼ Test data
★ Predictions on train data
✱ Predictions on test data

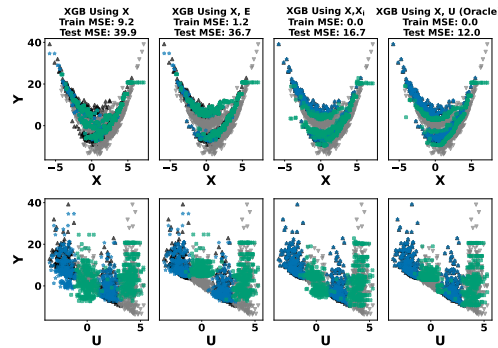


Figure D20: Performance of XGB on the low overlap non-linear setting, for sets of features: (i) X (ii) X and E (environment statistics) (iii) X and X_i (iv) X and U (Oracle).

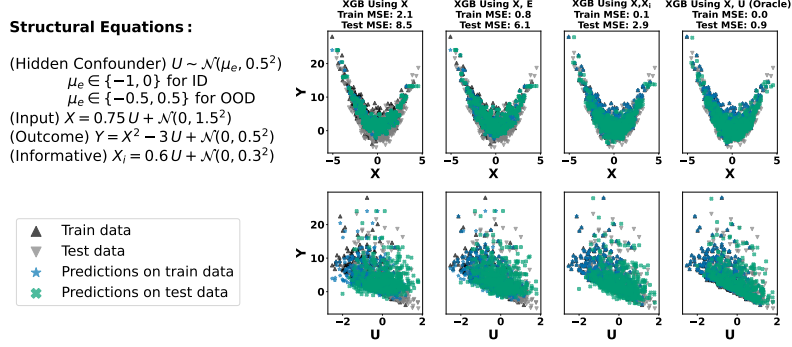


Figure D21: Performance of XGB on the low overlap non-linear setting, for sets of features: (i) X (ii) X and E (environment statistics) (iii) X and X_i (iv) X and U (Oracle).

data. To determine whether this baseline superiority stems from the $\mathbf{X} \rightarrow Y$ structure, we vary the causal-to-anti-causal ratio $\rho = |\mathbf{X}_C|/|\mathbf{X}_A| \in \{0.0, \dots, 1.0\}$ with $|\mathbf{X}| = 50$ and $\mathbf{X} = \mathbf{X}_C \cup \mathbf{X}_A$ (e.g., $\rho = 0.5$ gives $|\mathbf{X}_C| = |\mathbf{X}_A| = 25$). The baselines remain resilient even under $\mathbf{X}_C \rightarrow Y \rightarrow \mathbf{X}_A$ (Figure D22), achieving superior PMA-OOD scores [24] (fraction of maximum OOD accuracy), though no single method dominates universally (Figures D23, D24). We evaluate: (1) ERM-based methods (XGBoost [15], LightGBM [34], MLP, ResNet [27], SAINT [76], TabTransformer, NODE [61], FT-Transformer, ExpGrad [2]); (2) OOD generalization methods (IRM [6], IB-IRM, IB-ERM [30], CausIRL [17], DANN [3], MMD [43], DeepCORAL [81], V-REx [38], Domain Mixup [90, 91]); and (3) Robust optimization methods (DRO [42], GroupDRO [69], Label DRO, Adversarial Label DRO [94]).

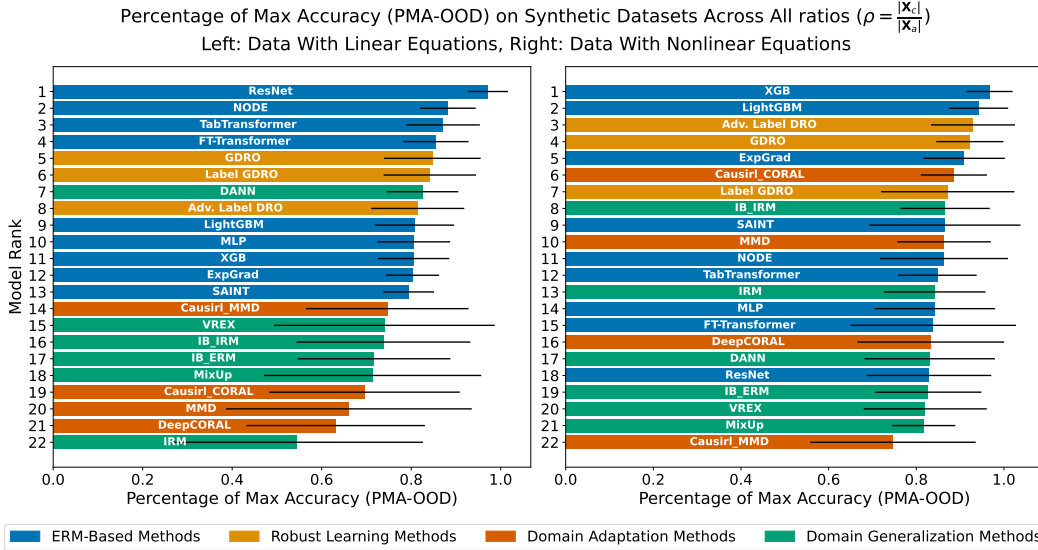


Figure D22: ERM-based methods achieve a higher percentage of maximum OOD (PMA-OOD) test accuracy on the synthetic datasets, mirroring the real-world results of [24].

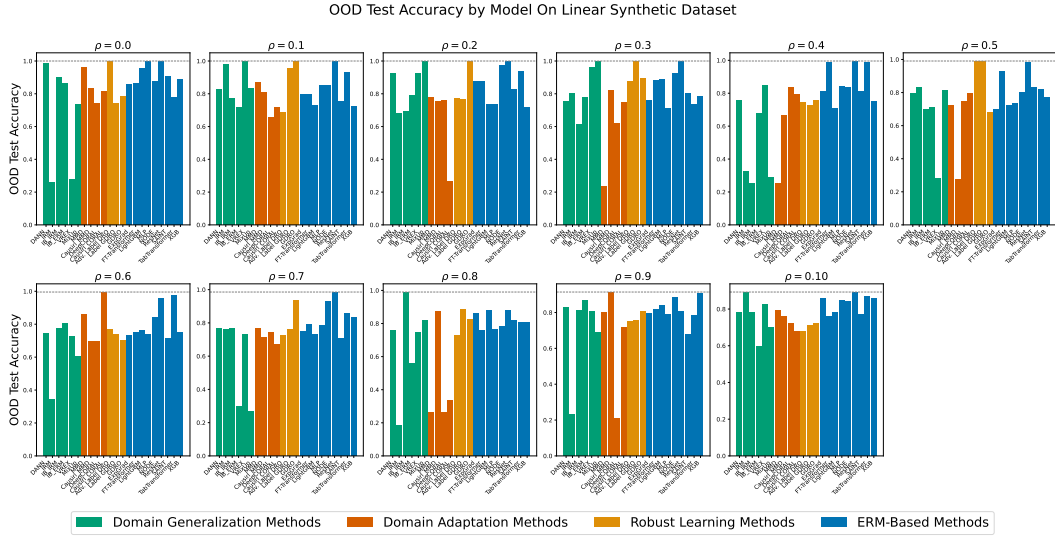


Figure D23: OOD test accuracy on the linear setting per model for different values of ρ .

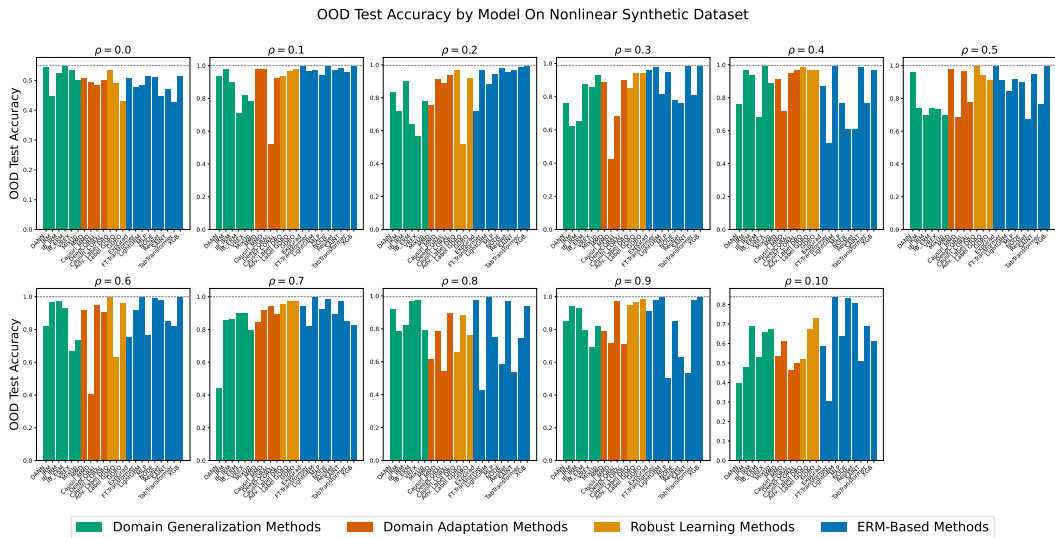


Figure D24: OOD test accuracy on the non-linear setting per model for different values of ρ

840 E Qualitative analysis of hidden confounding shift in real-world datasets

841 This section qualitatively examines hidden confounding shifts in real-world datasets. Tables E3–
 842 E18 list plausible hidden confounders that influence the observed covariates in several real-world
 843 datasets [57]; for further dataset details, see [24, 57]. We use GPT4o [1] for this task; the resulting
 844 list is not exhaustive, but it consistently identifies concrete unobserved confounders, reinforcing the
 845 evidence that hidden confounding shifts exist and matter across domains in real-world data.

846 **Prompt:** We query GPT4o for possible hidden confounders with the prompt below.

847 For a target variable, you will be given lists of causal, arguably
 848 causal, anti-causal, and spurious covariates. Also, you will be provided
 849 with environment variables such that different datasets are collected in
 850 environments induced by these environment variables. Provide possible
 851 hidden confounders that cause any of the target, causal, arguably causal,
 852 spurious, and anti-causal covariates. That is, the hidden confounders
 853 should influence the distribution of their children in different domains.
 854 Ensure hidden confounders are not a part of the given variables, and your
 855 explanation should include how hidden confounders influence their children
 856 in different environments.

Table E3: **Dataset:** Food stamps. **Target:** Food stamp reciprocity in the past year for households with child. **Environments:** regions in the United States.

Hidden confounder	Affected variables	Reason for confounding
Local economic conditions	Food stamp reciprocity, Age, Sex, Race, Marital status	Poor economic conditions can increase food insecurity (raising food stamp usage), while also influencing population demographics due to migration, employment patterns, and household formation.
State or regional public assistance policies	Food stamp reciprocity, Marital status, Number of children, Household income	Generous or restrictive assistance policies directly affect food stamp eligibility and indirectly influence household structure and income distribution.
Cultural attitudes towards welfare and public assistance	Food stamp reciprocity, Marital status, Disability, Race, Ethnicity	Social stigma or support for welfare affects both the uptake of food stamps and social norms around marriage, disability reporting, and identity categories.
Regional health care conditions	Food stamp reciprocity, Disability, Cognitive difficulty, Hearing difficulty, Vision difficulty	Regions with limited healthcare access may show higher disability prevalence and greater reliance on food stamps due to increased financial and care burdens.

Table E4: **Dataset:** Income. **Target:** Total person’s income $\geq 56K$ for employed adults. **Environments:** regions in the United States.

Hidden confounder	Affected variables	Reason for confounding
Local economic conditions	Income, Occupation, Educational attainment, Marital status	Stronger local economies offer better-paying jobs and education access, increasing income; they also influence occupational choices and household formation patterns.
State or regional tax and labor policies	Income, Class of worker, Educational attainment, Marital status	Tax incentives and labor protections affect wages and employment types, while also shaping decisions about education and family due to financial security.
Cost of living and regional affordability	Income, Occupation, Marital status, Educational attainment	Higher living costs necessitate higher incomes and may drive occupational or educational shifts; they also influence marriage or cohabitation decisions.
Cultural norms and regional economic history	Income, Educational attainment, Occupation	Cultural values and historical industry presence shape education levels and job opportunities, which in turn affect income distributions.

Table E5: **Dataset:** Public coverage. **Target:** Coverage of non-Medicare eligible low-income individuals. **Environments:** Disability statuses.

Hidden confounder	Affected variables	Reason for confounding
State or regional healthcare policies	Public health coverage, Marital status, Employment status, Income	Differences in Medicaid expansion and public insurance eligibility influence health coverage; these policies also affect economic stability, employment, and family dynamics.
Economic conditions and poverty levels	Public health coverage, Employment status, Income, Marital status	Poorer regions tend to have lower employment rates and incomes, which increase reliance on public health coverage and influence marriage and household composition.
Cultural attitudes towards disability and healthcare	Public health coverage, Disability status, Cognitive difficulty, Hearing difficulty, Vision difficulty	Stigma or support for disability and public care varies culturally, affecting both the reporting of disabilities and the likelihood of seeking or receiving public coverage.
Urban vs. rural divide	Public health coverage, Educational attainment, Marital status, Occupation	Urban areas typically offer better healthcare access, education, and jobs, all of which affect coverage likelihood and socioeconomic indicators.

Table E6: **Dataset:** Unemployment. **Target:** Classify whether a person is unemployed. **Environments:** Educational attainments.

Hidden confounder	Affected variables	Reason for confounding
Local economic conditions	Employment status, Occupation, Marital status, Mobility status	Regional economic strength affects job availability and unemployment rates; it also shapes occupation types, migration decisions, and household stability.
State or regional labor market policies	Employment status, Occupation, Educational attainment, Marital status	Labor regulations and unemployment benefits vary by region, affecting hiring practices, education incentives, and family structures.
Cultural and social norms around employment	Employment status, Disability status, Marital status	Cultural attitudes toward work and dependency influence unemployment reporting and societal roles around disability and family responsibilities.
Health and disability status	Employment status, Disability status, Cognitive difficulty, Hearing difficulty, Vision difficulty	Poor health or disabilities reduce the ability to work, directly increasing unemployment and shaping associated health-related variables.

Table E7: **Dataset:** Voting. **Target:** Classify whether a person voted in the U.S. presidential election. **Environments:** United States census regions.

Hidden confounder	Affected variables	Reason for confounding
Political, family, and peer influences	Voted in national election, Party identification, Political participation	Social networks shape political ideology and engagement, influencing both voting likelihood and party alignment.
Media consumption habits	Voted in national election, Party identification, Political knowledge, Voting behavior	Media exposure affects awareness of political issues and biases, influencing party affiliation, political knowledge, and voting participation.
Social capital	Voted in national election, Political participation, Interest in elections	Strong community ties and civic networks increase political interest and participation, leading to higher voter turnout.
Civic education and political engagement programs	Voted in national election, Interest in elections, Political knowledge	Educational programs raise political awareness and civic responsibility, influencing both knowledge levels and voting decisions.
Historical and cultural context	Voted in national election, Party identification, Interest in elections	Historical events and regional political culture affect interest in elections and party alignment, which in turn influence voting behavior.

Table E8: **Dataset:** Hypertension. **Target:** Whether a person has hypertension. **Environments:** Body Mass Index (BMI) values.

Hidden confounder	Affected variables	Reason for confounding
Socioeconomic status	Income, Employment status, Smoking habits, Alcohol consumption, Physical activity, Healthcare access, Medical cost, High blood pressure diagnosis	Lower socioeconomic status reduces access to healthcare and healthy lifestyle options, leading to poor diet, limited activity, and delayed diagnosis, which jointly influence both BMI and hypertension risk.
Access to healthcare services	Healthcare access, Medical costs, Smoking habits, Alcohol consumption, Physical activity, High blood pressure diagnosis	Limited healthcare access results in underdiagnosis and unmanaged hypertension, while also affecting lifestyle choices that vary with BMI, confounding the relationship between BMI and hypertension.
Genetic predisposition	Age group, Race, Sex, Smoking habits, Diabetes, High blood pressure diagnosis	Genetic risk factors for hypertension may co-vary with demographic attributes and influence both hypertension prevalence and BMI distribution across subpopulations.
Psychosocial stress	Smoking habits, Alcohol consumption, Physical activity, High blood pressure diagnosis, Diabetes, Age group	Chronic stress alters behavior (e.g., smoking, inactivity) and physiological responses, contributing to both increased BMI and elevated blood pressure, thus confounding the BMI–hypertension link.
Environmental factors	Physical activity, Diet, Smoking habits, High blood pressure diagnosis, Diabetes, BMI category	Living environments affect access to recreational spaces, food quality, and pollution exposure, influencing both BMI and hypertension risks. These vary across BMI categories, creating confounding.
Dietary habits beyond fruits and vegetables	Alcohol consumption, Smoking habits, Physical activity, High blood pressure diagnosis, BMI category	High-sodium or processed-food diets raise both BMI and hypertension risk. Variation in such unmeasured dietary habits across BMI categories creates spurious associations with hypertension.

Table E9: **Dataset:** College Scorecard. **Target:** Predict completion rate for first-time, full-time students at four-year institutions. **Environments:** Based on Carnegie Classifications.

Hidden confounder	Affected variables	Reason for confounding
Institutional funding and re-sources	Accreditor, Control of institution, Highest degree awarded, In-state tuition, Out-of-state tuition, Cost of attendance, SAT scores	Wealthier institutions can offer better academic support, facilities, and programs, leading to higher completion rates and more selective admission profiles. This varies across Carnegie classifications.
Regional socio-economic factors	Region, Poverty rate, Unemployment rate, SAT scores	Economic conditions across regions affect affordability, student preparedness, and institutional support levels, all influencing both enrollment outcomes and graduation likelihood.
Demographic factors	HBCU flag, Federal loan recipient rate, Pell grant recipient rate, ACT scores, Undergraduate enrollment	Student demographics shape financial aid needs, academic preparation, and graduation rates. Their effect differs by institution type and selectivity under Carnegie categories.
Community support and engagement	Distance-education flag, Federal loan recipient rate, Pell grant recipient rate, SAT scores, Undergraduate enrollment	Supportive institutional communities improve retention and completion. Variation in engagement across institution types and student aid profiles induces confounding.
Admission selectivity	Admission rate, SAT (reading/math) midpoints, ACT midpoint, Undergraduate enrollment	Selective admissions correlate with better-prepared students and higher completion rates, and vary with institutional prestige and classification.
State and local policies	Region, Poverty rate, Unemployment rate, Cost of attendance	Differences in education funding and public policy affect cost structures and completion outcomes, interacting with institutional classification and regional demographics.

Table E10: **Dataset:** ASSISTments. **Target:** Predict whether a student solves a problem correctly on the first attempt in an online learning tool. **Environments:** Different schools.

Hidden confounder	Affected variables	Reason for confounding
Institutional teaching quality	Hint count, Attempt count, Skill ID, Problem type, Tutor mode, Position, Type, First action, Milliseconds to first response, Overlap time, Average confidence	Variation in instructional quality and pedagogy across schools affects how effectively students engage with content, leading to differences in problem-solving strategies, response behavior, and emotional states.
Student motivation	Hint count, Attempt count, Skill ID, Problem type, First action, Milliseconds to first response, Overlap time, Average confidence	Differences in intrinsic motivation across schools influence students' willingness to persevere, seek help, or give up quickly, affecting interaction and performance.
Classroom environment	Hint count, Attempt count, Tutor mode, Position, Type, First action, Average confidence	Peer dynamics, classroom culture, and noise levels vary by school and affect how confidently and independently students solve problems.
School technology infrastructure	Hint count, Attempt count, Tutor mode, Position, Type, First action, Milliseconds to first response, Overlap time, Average confidence	Access to reliable devices and fast internet differs by school, influencing response time, tool usage, and student experience.
Teacher-student interaction	Hint count, Attempt count, Tutor mode, Position, Type, First action, Milliseconds to first response, Overlap time, Average confidence	The level of teacher guidance and feedback shapes how much support students require during problem-solving, affecting engagement and confidence.
Previous academic performance	Hint count, Attempt count, Skill ID, Problem type, First action, Average confidence	Students' prior achievement affects how easily they solve problems, their need for assistance, and their confidence, all of which vary across schools.

Table E11: **Dataset:** ICU. **Target:** Predict whether the patient will stay in the ICU for longer than 3 days. **Environments:** Insurance types.

Hidden confounder	Affected variables	Reason for confounding
Socioeconomic status (SES)	Age, Gender, Ethnicity, Height, Weight, Bicarbonate, CO ₂ , pCO ₂ , pO ₂ , Lactate, Sodium, Hemoglobin, Oxygen saturation, Respiratory rate, etc.	SES influences access to healthcare, preventive services, and overall health status. Differences in SES across insurance types lead to variability in pre-ICU health, physiological indicators, and ICU stay duration.
Hospital resources and care quality	Age, Gender, Ethnicity, Height, Weight, Bicarbonate, CO ₂ , pCO ₂ , pO ₂ , Lactate, Sodium, Hemoglobin, Oxygen saturation, Respiratory rate, Heart rate, etc.	Hospital infrastructure and care standards affect monitoring, intervention speed, and clinical decisions. These factors vary by insurance coverage and influence ICU outcomes and vitals.
Comorbidities	Age, Gender, Ethnicity, Height, Weight, Bicarbonate, CO ₂ , pCO ₂ , pO ₂ , Lactate, Sodium, Hemoglobin, Oxygen saturation, Respiratory rate, Heart rate, etc.	Presence of chronic conditions (e.g., diabetes, cardiovascular disease) affects both the need for prolonged ICU care and physiological measurements. The distribution of comorbidities differs across insurance types.
Insurance-related treatment variability	Age, Gender, Ethnicity, Height, Weight, Bicarbonate, CO ₂ , pCO ₂ , pO ₂ , Lactate, Sodium, Hemoglobin, Oxygen saturation, Respiratory rate, etc.	Differences in treatment timing, intensity, and access to specialists based on insurance policies affect ICU stay duration and clinical metrics.
Genetic factors	Age, Gender, Ethnicity, Height, Weight, Bicarbonate, CO ₂ , pCO ₂ , pO ₂ , Lactate, Sodium, Hemoglobin, etc.	Inherited traits influence predisposition to organ failure, metabolic responses, and recovery trajectories. These effects are partially mediated by ethnicity and age distributions, which vary across insurance groups.

Table E12: **Dataset:** Hospital mortality. **Target:** Classify whether an ICU patient expires in the hospital during their current visit. **Environments:** Insurance types.

Hidden confounder	Affected variables	Reason for confounding
Socioeconomic status (SES)	Age, Gender, Ethnicity, Height, Weight, Bicarbonate, Lactate, Sodium, Hemoglobin, Oxygen saturation, Respiratory rate, Systolic blood pressure, White blood cell count, etc.	SES shapes access to timely and high-quality care, preventive services, and general health status. Patients with higher SES often have better insurance and outcomes, leading to confounding with mortality risk.
Comorbidities	Age, Gender, Ethnicity, Height, Weight, Bicarbonate, CO ₂ , pCO ₂ , pO ₂ , Lactate, Sodium, Hemoglobin, Oxygen saturation, Respiratory rate, Heart rate, Systolic blood pressure, etc.	Pre-existing conditions such as diabetes or heart disease increase mortality risk and influence physiological features. Their prevalence differs by insurance type, creating confounding.
Hospital resources and care quality	Age, Gender, Ethnicity, Height, Weight, Bicarbonate, CO ₂ , pCO ₂ , pO ₂ , Lactate, Sodium, Hemoglobin, Oxygen saturation, Respiratory rate, Heart rate, Systolic blood pressure, White blood cell count, etc.	Access to advanced treatments, trained staff, and timely interventions influences survival rates. These factors correlate with insurance coverage, confounding mortality outcomes.
Genetic factors	Age, Gender, Ethnicity, Height, Weight, Bicarbonate, CO ₂ , pCO ₂ , pO ₂ , Lactate, Sodium, Hemoglobin, Oxygen saturation, Respiratory rate, Heart rate, etc.	Genetic predispositions affect disease susceptibility and treatment responses. Variations in genetic risk factors may correlate with demographic traits across insurance types.
Lifestyle and behavioral factors	Age, Gender, Ethnicity, Height, Weight, Bicarbonate, CO ₂ , pCO ₂ , pO ₂ , Lactate, Sodium, Hemoglobin, Oxygen saturation, Respiratory rate, Heart rate, Temperature, Systolic blood pressure, White blood cell count, etc.	Behaviors such as smoking, diet, and physical activity affect long-term health and mortality risk. These behaviors vary systematically with SES and insurance coverage, influencing both target and physiological features.

Table E13: **Dataset:** Childhood lead. **Target:** Predict blood lead levels above CDC blood level reference value. **Environments:** Poverty-income ratios.

Hidden confounder	Affected variables	Reason for confounding
Environmental Exposure	Blood lead levels, Country of birth, Age, Race and Hispanic origin	Environmental exposure to lead influences blood lead levels, and this varies significantly across socio-economic groups. People in lower PIR groups are more likely to live in areas with higher lead contamination, which contributes to higher blood lead levels. Furthermore, environmental factors may affect the demographic distribution (e.g., country of birth, race).
Access to Healthcare	Blood lead levels, Age, Gender, Race and Hispanic origin, Marital status, Education	Limited access to healthcare, especially in lower PIR groups, means fewer opportunities for detection and treatment of lead poisoning. This results in higher blood lead levels, with disparities also influencing demographic variables like age, gender, and education. Additionally, healthcare access varies by insurance and socio-economic status, further confounding the relationships.
Diet and Nutrition	Blood lead levels, Age, Gender, Race and Hispanic origin	Dietary factors, such as poor nutrition in lower PIR groups, can exacerbate lead absorption. Malnutrition increases the body's susceptibility to lead poisoning, raising blood lead levels. In contrast, higher PIR groups may have better access to nutritious foods, lowering lead absorption, thus creating a confounding effect in how socio-economic status and race influence lead toxicity.
Housing Conditions	Blood lead levels, Country of birth, Race and Hispanic origin, Marital status	Older housing conditions, which are more prevalent in lower PIR groups, contribute significantly to elevated lead exposure (e.g., lead paint, poor plumbing). These living conditions directly influence blood lead levels and can also correlate with demographic factors like country of birth, race, and marital status. This introduces confounding, as socio-economic status impacts both exposure and the demographics of affected individuals.
Occupation	Blood lead levels, Age, Race and Hispanic origin, Marital status, Education	Certain occupations, which are more common among lower PIR groups, involve higher lead exposure (e.g., construction, manufacturing). Occupational lead exposure directly impacts blood lead levels and is often correlated with education, marital status, and socio-economic status. The varying prevalence of lead exposure by occupation introduces confounding, especially across different PIR groups.

Table E14: **Dataset:** Diabetes. **Target:** Predict diabetes. **Environments:** Preferred race categories.

Hidden confounder	Affected variables	Reason for confounding
Genetic predisposition	Diabetes, BMI, High blood pressure, High blood cholesterol	Genetic factors and family history contribute to both the onset of diabetes and comorbid conditions like obesity, hypertension, and high cholesterol. These genetic predispositions can make individuals more susceptible to diabetes, leading to confounding as they correlate with other health indicators.
Access to healthcare	Diabetes, Physical health, BMI, Healthcare coverage, Health checkups	Limited or unequal access to healthcare, especially in marginalized racial groups, leads to disparities in diabetes diagnosis, management, and comorbidity treatment. It also influences the frequency of health checkups and access to medications, which can confound the relationship between diabetes status and other health metrics.
Dietary habits and food availability	Diabetes, BMI, Physical health, Alcohol consumption, Fruit and vegetable intake	Dietary habits, often shaped by socioeconomic status and local food environments, influence weight, health behaviors (such as alcohol consumption), and diabetes risk. People in lower socioeconomic strata may have limited access to healthy food options, leading to higher BMI and increased diabetes risk, creating confounding effects on health outcomes.
Psychosocial stress and mental health factors	Diabetes, Mental health, Physical health, BMI, Physical activity, Doctor visits	Chronic stress and mental health issues, often higher in marginalized groups, contribute to diabetes development and complicate its management. These factors also affect physical health (e.g., weight gain due to stress) and health-seeking behaviors (e.g., fewer doctor visits), leading to confounding by influencing both diabetes risk and its associated variables.
Socioeconomic status beyond income	Diabetes, Income, Physical health, Healthcare coverage, Education level	Socioeconomic factors, such as occupation, education, and neighborhood wealth, influence access to healthcare, nutrition, and overall health behaviors. These factors can confound the relationships between diabetes and other socio-economic variables like income and education, as they shape opportunities for prevention and treatment.

Table E15: **Dataset:** Sepsis. **Target:** Predict, from a set of fine-grained ICU data, whether a patient will experience sepsis onset within the next 6 hours. **Environments:** Lengths of ICU stay.

Hidden confounder	Affected variables	Reason for confounding
Infection prevalence in ICU	SepsisLabel, Temperature (Temp), Leukocyte count (WBC), Heart rate (HR), Blood urea nitrogen (BUN)	Higher infection rates in certain ICU units can lead to a higher probability of sepsis onset (SepsisLabel). These infection rates influence biomarkers such as WBC, HR, and BUN, creating confounding because the unit's infection environment affects both the likelihood of sepsis and the observed clinical measures.
Quality of ICU Care	SepsisLabel, Fibrinogen concentration (Fibrinogen), Leukocyte count (WBC), Platelet count (Platelets)	Higher-quality care in certain ICUs may lead to earlier identification and treatment of sepsis, resulting in more accurate SepsisLabel predictions. Additionally, better care could affect biomarkers like fibrinogen, WBC, and platelets, which are critical in sepsis detection and progression, thereby confounding the relationships between these variables and the outcome.
Severity of underlying conditions	SepsisLabel, Blood urea nitrogen (BUN), Creatinine, Lactate, Calcium	Patients with severe chronic conditions (e.g., kidney disease, cardiovascular issues) are at a higher risk of sepsis and may show abnormal levels in biomarkers like BUN, creatinine, lactate, and calcium. These underlying conditions contribute to the SepsisLabel outcome and confound the relationship between the biomarkers and the likelihood of sepsis, varying across ICU units depending on patient population.
Patient's socio-economic status	Age (Age), Gender (Gender), Leukocyte count (WBC), Fibrinogen concentration (Fibrinogen), Platelet count (Platelets)	Socio-economic factors, such as access to healthcare, can influence both the likelihood of sepsis and the observed clinical biomarkers. For example, patients from lower socio-economic backgrounds may have delayed hospitalizations or inadequate care, which affects both SepsisLabel and the progression of sepsis as indicated by WBC, fibrinogen, and platelet levels.
Hospital-specific protocols and treatment guidelines	SepsisLabel, Lactate, Glucose, Creatinine	Differences in hospital protocols for sepsis treatment, such as timing of interventions and choice of sepsis bundles, can affect both the SepsisLabel and biomarkers like lactate, glucose, and creatinine. These protocols lead to variability in how sepsis is diagnosed and treated across different hospitals and ICU units, confounding the relationship between biomarkers and sepsis outcomes.

Table E16: **Dataset:** Hospital readmission. **Target:** Predict whether a diabetic patient is readmitted to the hospital within 30 days of their initial release. **Environments:** Admission sources.

Hidden confounder	Affected variables	Reason for confounding
Socio-economic status (SES)	Race, Gender, Age, Payer code, Medical specialty, Number of outpatient visits, Number of emergency visits, Number of inpatient visits, Diabetes medication prescribed	SES influences access to healthcare, patient demographics, and chronic disease rates. It can also determine the type of care received based on admission source (e.g., emergency department vs. outpatient settings). Differences in healthcare access, such as availability of medications, may impact readmission rates and associated variables like outpatient visits and prescribed medication.
Severity of illness	Primary diagnosis, Secondary diagnosis, Number of diagnoses, Discharge type, Medication changes (e.g., Insulin, Glipizide)	More severe illness increases the likelihood of readmission and influences the complexity of diagnoses and the treatments administered. The severity of illness may differ based on the admission source (e.g., emergency versus outpatient), impacting the number and type of diagnoses and treatments prescribed at discharge, affecting readmission likelihood.
Access to healthcare resources	Time in hospital, Discharge disposition, Number of procedures, Number of medications, Number of lab tests	Access to healthcare resources (e.g., time in hospital, availability of procedures and medications) influences treatment decisions and outcomes. Different admission sources may have varying levels of available resources, leading to different lengths of stay, the types of procedures performed, and overall treatment quality, which can affect the likelihood of readmission.
Patient's adherence to medication	Change in medications, Diabetes medication prescribed, Number of outpatient visits, Number of emergency visits	Adherence to prescribed medications is often influenced by socio-economic status, which can vary depending on admission source. Non-adherence may lead to medication changes, affecting diabetes control and subsequent readmission risk. The number of outpatient and emergency visits can also reflect how well a patient manages their diabetes and the likelihood of complications.
Hospital-specific protocols	Time in hospital, Number of diagnoses, Discharge disposition, Readmitted	Different hospitals and healthcare systems implement various protocols for discharge planning and readmission prevention, which can affect readmission rates. These protocols may vary by admission source, where patients admitted via the emergency department may receive different follow-up instructions and care than those admitted through other channels.

Table E17: **Dataset:** MEPS. **Target:** Measure of health care utilization. **Environments:** Insurance types.

Hidden confounder	Affected variables	Reason for confounding
Socioeconomic status (SES)	Years of education, Employment status, Hourly wage, Paid sick leave, Paid leave to visit doctor, Family size, Insurance coverage, Healthcare utilization	SES can influence access to healthcare services, insurance coverage, and the ability to use medical services. Insurance types often correlate with SES levels, where individuals with lower SES may be more likely to be insured by government programs (e.g., Medicaid), which in turn impacts healthcare utilization patterns across different SES groups.
Healthcare access	Paid sick leave, Insurance coverage, Employer offers health insurance, Healthcare utilization	Limited access to healthcare, such as lack of insurance or paid sick leave, directly impacts healthcare utilization. The type of insurance a person holds is often tied to access to various medical services. The level of coverage and accessibility differs across insurance types, influencing healthcare behaviors such as whether a patient can afford and utilize healthcare services.
Health behaviors	Perceived health status, Asthma medications, Limitations in physical functioning, Healthcare utilization	Lifestyle factors like smoking, alcohol consumption, and physical activity can directly affect health status and healthcare needs. Health behaviors differ across groups with different insurance types, and these behaviors contribute to healthcare utilization. Insurance coverage can also be influenced by perceived health status, which varies across insured groups, affecting utilization of medical services.
Chronic health conditions	Asthma medications, Perceived health status, Limitations in physical functioning, Healthcare utilization	Chronic conditions like asthma or diabetes increase the need for healthcare services, leading to higher healthcare utilization. People with chronic conditions are more likely to be covered by Medicare or Medicaid, which influences the type of insurance they hold and impacts their healthcare utilization and associated features (e.g., medications and physical limitations).
Regional healthcare infrastructure	Region, Family size, Healthcare utilization, Paid sick leave, Paid leave to visit doctor, Employment status	The quality and availability of healthcare infrastructure vary across regions, which can impact healthcare utilization. In underserved regions, people may be more reliant on public insurance options like Medicaid, affecting healthcare access and behaviors. The regional differences in healthcare systems can lead to disparities in access to medical services, insurance coverage, and utilization of healthcare.

Table E18: **Dataset:** Poverty. **Target:** Predict household income-to-poverty ratio. **Environments:** Citizenship statuses.

Hidden confounder	Affected variables	Reason for confounding
Social capital	Income, Unemployment compensation, Disability benefits, Family size, Housing conditions, Household income-to-poverty ratio	Stronger social support networks can improve access to income, benefits, and resources like unemployment or disability compensation. Social capital can vary based on citizenship status, which in turn influences income, housing conditions, and eligibility for benefits, confounding the relationship between household income and poverty ratios.
Workplace discrimination or bias	Household income, Worker's compensation, Unemployment compensation, Pension, Family income, Disability benefits	Discrimination in the workplace can limit opportunities for higher wages, pensions, or disability benefits, especially for marginalized groups (e.g., racial minorities, non-citizens). This bias varies by citizenship status and leads to biased associations between income, benefits, and poverty levels.
Healthcare access	Household income, Health insurance premiums, Medical expenses, Disability benefits, Family size, Health conditions, Medicaid/Medicare assistance	Limited access to healthcare, particularly for non-citizens, can lead to higher medical expenses and greater reliance on public health assistance. Citizenship status directly impacts eligibility for public healthcare programs, confounding relationships between medical expenses and income, and affecting poverty ratios.
Access to education and skills development	Educational attainment, Household income, Employment status, Unemployment compensation, Income from assistance, Savings	Disparities in educational opportunities, often linked to citizenship status, influence income potential and eligibility for government assistance. These disparities, in turn, affect the household income-to-poverty ratio and employment outcomes, creating confounding relationships between education and income.
Cultural factors	Family size, Household income, Savings, Disability benefits, Healthcare utilization, Income assistance, Living arrangements	Cultural factors influence financial management, family support, and the use of social programs. These factors can vary across citizenship statuses, leading to differences in how household income and assistance are distributed, and thus confounding the relationship between income, benefits, and financial behavior.
Housing market and rent conditions	Housing ownership, Housing costs, Family size, Household income, Rent payments, Social security benefits, Medical aid	Housing market conditions, especially rent disparities, can create financial strain and impact the household income-to-poverty ratio. Citizenship status often influences eligibility for housing assistance and rent subsidies, which confounds the relationship between housing costs and income, particularly in areas with significant immigrant populations.